The New Work of Generative AI

Evidence from 18 Million Duty Statements in US Job Postings

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Abstract

Whether generative AI raises or lowers the demand for labour depends on the new work it creates. I measure that new work directly in 18.6 million duty statements from US job postings, 2019–2026, drawn from a new dataset built from applicant-tracking-system pages archived by Common Crawl. A duty is new when nothing in early-2019 postings is close to it. In the third of occupations that use AI most, genuinely new work rose from about one in every 100 duty lines in 2022 to nearly five in 2026, four-fifths of it work on AI itself. Its mix is shifting from building AI systems to working with them: using AI tools and driving their adoption. Genuinely new tasks that are not about AI remain rare, and half of the new AI duties have no related task in their occupation's O*NET list. The pay premium for AI duties is mostly sorting.

JEL codes: J23, J24, O33. **Keywords:** generative AI, new work, job postings, tasks, reinstatement.

1 Introduction

New technologies displace workers from some tasks and create new tasks for them. Whether a technology raises the demand for labour on net depends on the second effect, which Acemoglu and Restrepo (2018, 2019) call reinstatement. Historically, new work has appeared where innovations complement the tasks that workers already do (Lin, 2011; Autor et al., 2024). For generative AI the evidence so far is on the other side of the ledger: which occupations and tasks the technology can perform (Felten et al., 2021; Webb, 2020; Eloundou et al., 2024; Tomlinson et al., 2025), which ones people use it for (Handa et al., 2025; Massenkoff and McCrory, 2026), and whether employment in exposed occupations has fallen (Brynjolfsson et al., 2025; Hampole et al., 2025; Humlum and Vestergaard, 2025). What new work generative AI creates, and for whom, has not been measured directly in the content of jobs.

This paper measures it in job postings. A posting lists what an employer asks a new hire to do, one duty at a time. I compare every duty statement in US postings from 2019 to 2026 with the duty statements of early 2019, and call a statement new when nothing in the 2019 postings is close to it. A language model then reads the new statements: whether they describe work on AI, which side of AI they are on, and whether the rest are genuinely new tasks or old duties in new words. To

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do this I build a new dataset of US job postings from the job pages of applicant tracking systems (ATS) archived by Common Crawl.

Three facts stand out. First, new work has grown where AI is used. In the third of occupations with the most AI use, duty lines that are genuinely new work rose from about one in every 100 in 2022 to nearly five in 2026; in the third with the least use they remained below one in 200. Second, most of that new work is work on AI, and its mix is shifting. Building, deploying and selling AI systems made up most AI duties until 2023. Since then, duties that use, direct, review or drive the adoption of AI tools have grown from 5% of AI duty lines (2022) to 24% (2026), and in health care, finance and retail they are now as common as building AI. Third, genuinely new tasks that are not about AI remain rare. Most of the apparently new non-AI lines are variants or re-wordings of old duties; most rising non-AI themes are templates and new phrasing; and the new job titles that carry the most new duties are AI titles. The new AI duties are also largely missing from the official list of tasks: half of them have no task of even related work in their occupation's O*NET list, against one in eight continuing duties and one in three new duties that are not about AI, so measures built on that list miss much of the new work.

Two further results describe the new work. Postings with AI duties advertise higher pay, mostly because they are at high-paying firms and, for working with AI, in more senior roles: within the same title at the same firm, the premium is about 3%. And across occupations, working-with-AI duties rose from late 2024, more in the occupations that use AI more, after flat trends before.

So far, then, generative AI has reinstated labour mainly through tasks that operate the technology itself. That is the pattern of earlier general-purpose technologies (Autor et al., 2024), and the shift from building AI to working with it is what one would expect as a technology diffuses from its producers to its users. Whether these tasks last, or are themselves the next to be automated, these data cannot yet say.

The paper also contributes data. Research on what job postings ask for relies mostly on commercial data that are costly and cannot be redistributed (Hershbein and Kahn, 2018; Deming and Kahn, 2018; Atalay et al., 2020; Acemoglu et al., 2022). The postings used here come from a public archive, and I will release the code that builds them together with the derived data: each posting's title, employer, dates, location, pay, occupation code and weights, the classification of every line of its text, and the address of the archived page, from which anyone can rebuild the text. Although these data are a thin and selective slice of US hiring, tilted towards professional and technical jobs, they track official statistics and commercial postings closely: across states and detailed occupations, in posted pay, in the trends of the shares that state pay or offer remote work, and through known shocks such as Colorado's pay-transparency law and the move to remote work during the pandemic (Section 2 and Appendix A).

The closest work uses postings to study generative AI. Chen et al. (2024) count, in Lightcast postings to 2024, the skills from Lightcast's taxonomy that are new to a firm and occupation, and find more of them in occupations where AI augments work; I classify the duty text itself and ask whether the new lines are new work, which most of the apparently new non-AI lines are not. Wang et al. (2026) score the exposure of tasks extracted from Lightcast postings and decompose its decline into hiring shifts and changes within jobs, and Shukla (2026) maps 288 million postings line by line to O*NET tasks to measure the work AI cannot do; both measure the AI exposure of the tasks postings list, where this paper is about the duties that are new. Hosseini and Lichtinger (2026) identify firms that adopt generative AI by their first posting for workers who integrate it, and find that junior employment falls at adopters; Chandar and Klein Teeselink (2026) find similar shifts

across 41 countries. On new work, Marguerit (2025) measures it with new O*NET job titles before generative AI, and Humlum and Vestergaard (2025) find in Danish surveys that chatbots create some new tasks; Acemoglu et al. (2022) study earlier AI in vacancies. Studies of the job ladder ask whether AI reduces junior hiring (Brynjolfsson et al., 2025; Hosseini and Lichtinger, 2026; Iscenko and Curto Millet, 2026). Meisenbacher et al. (2025) release aggregate O*NET features from postings on public job banks; the data here are at the level of the posting and the duty line.

Section 2 describes the data and Section 3 the measures. Section 4 presents the results, Section 5 the limitations, and Section 6 concludes.

2 Data

2.1 Postings from applicant tracking systems

Large employers publish their job adverts through applicant tracking systems: software that hosts the advert and collects applications, such as Greenhouse, iCIMS, Workday or Lever. An ATS can be detected for 97% of the Fortune 500 (Purcell, 2026). ATS job pages sit at predictable web addresses, and Common Crawl, a public archive of the web, has captured many of them since 2017 (Common Crawl Foundation, 2026). I start from 20 widely used ATS platforms. Eighteen of them have enough archived job pages to test: for each, I fetched 20 random archived job pages in each of four years (2018, 2021, 2024 and 2026) and checked what could be extracted. I keep the 14 platforms whose pages reliably give at least a title and a location, and query the archive's index for their job-page addresses in all 96 crawls from January 2017 to September 2026. They hold 84–95% of the US job pages that the archive holds for the 20 platforms (Table A1).

Each page is parsed twice, once for structured data (a schema.org `JobPosting` object) and once with a template for the platform, and each field comes from the structured data where it exists. The 10.5 million captured pages collapse to 5.2 million jobs by platform, employer and job identifier. Employers often repost an advert under a new identifier, so jobs with the same title, location and text are merged; the merge removes 7.9% of US jobs. A posting is called US from its location or, failing that, from the page metadata, the text, its other captures or the employer's other postings. The result is 3,341,872 US postings from 52,473 employers (Table A2). Each posting has its title, employer, location, pay where stated, text, the dates it was posted, first seen and last seen, and a code in the 2018 Standard Occupational Classification. I date postings by the crawl that first captured them, a median of 48 days after the stated date of posting.

Occupation codes come from a linear text classifier trained on postings coded by a language model (Claude Sonnet 5) and on postings whose titles match O*NET titles. On held-out postings it gives the same detailed code as the language model for 58% of postings and the same major group for 76%; two language models coding the same postings agree on 72% and 89%, which bounds what agreement with either can show. Every posting's detailed code places it in a third of occupations by AI use and gives it an O*NET task list; the occupation-level design and the themes use only codes the classifier is confident about, which cover 65% of postings at 73% detailed agreement (Appendix A.3).

The postings are a thin and selective slice of hiring. In 2024 they hold 0.5 postings for every 100 hires in the Job Openings and Labor Turnover Survey; in 2024–25, from 2.8 per 100 hires in computer occupations to 0.08 in construction. The slice is thin for three reasons: Common Crawl archives a sample of the web in each crawl, so it captures only a fraction of each platform's job

pages; the platforms cover part of the market, and Workday's pages carry text only from 2022; and many employers, notably school districts, public employers and health systems, post elsewhere. All online postings over-represent professional work: to match the occupation mix of employment, 34% of these postings would have to change major group, against 21% of Lightcast's commercial postings (see also Hershbein and Kahn, 2018).

Within that slice the content is sound (Figure A1). State shares correlate with employment shares at 0.92–0.98 a year; across detailed occupations, log posting shares correlate with Lightcast's at 0.74, and median posted pay with the occupation's median wage at 0.89; titles agree with an independent parse of the same pages for 99.9% of postings. Over time, the shares of postings that state pay or offer remote work follow the trends of Indeed's and Lightcast's series (correlations in levels of 0.85–0.98), though not their quarter-to-quarter changes, and the remote share runs above both from 2024. They move with known events: when Colorado required pay ranges in 2021, the share of its postings with pay rose from 10% to 55%, and the share offering remote work rose from 3% to 13% between early 2020 and late 2021. Appendix A reports these benchmarks and the data's known gaps. The sample over-represents computer and business occupations, so I report results by occupation group and check the aggregates with representativeness weights.

2.2 Duty statements

The unit of measurement is a duty statement: a sentence in which the employer lists something the new hire will do. A text layer restores the line structure of descriptions published as flat text (every Workday description, and some others), types each section header, splits the text into sentences and classifies each sentence as a duty, a requirement, a statement about the role, the company, pay and conditions, or process and legal text. Duties are further classed as specific tasks, broad responsibilities or generic statements ("other duties as assigned"), and generic ones are dropped. The classifier is trained on 1,200 postings whose every sentence Claude Sonnet 5.5 labelled while reading the whole advert; against those labels it finds duties with precision 0.92 and recall 0.94, and precision is above 0.89 in every year (Table A7) and at least 0.88 on every platform. A duty line is one duty sentence of one posting. Duty statements are close to the task statements of O*NET but not the same: a duty can name a tool, a setting or a standard, and a few lines that are not duties, mostly statements about the role, pass the classifier; the reading of the new lines (Section 3.3) flags them.

2.3 Analysis sample

I use postings first seen from 2019 to the third quarter of 2026. The crawls of 2017–18 captured some platforms at several times their later rate, so this paper does not use them; the last such crawl, in January 2019, falls in the reference half-year. I keep postings in English with a description of at least 200 characters, drop bulk posters (employer-years with at least 1,000 postings under at most 50 titles), keep one posting per multi-location advert, and keep postings with at least one duty line. The sample has 1,652,760 postings from 48,304 employer accounts and 18.6 million duty lines, of which 10.2 million are distinct sentences (Table 1).

Three further fields come from language-model labels extended by classifiers. Seniority comes from the job title: a model trained on 30,000 titles labelled by Sonnet 5.5 assigns one of eight levels from intern to executive, agreeing with the labels on 88% of held-out titles. Industry is a two-digit NAICS sector for each firm with at least three postings, coded by Sonnet 5.5 from the firm's name,

titles and occupation mix, and recoded from its company-description sentences where Sonnet was not confident; it covers 99% of postings. Firms that moved between platforms are linked by name, which joins 48,304 employer accounts into 45,678 firms.

Table 1: Analysis sample by first-seen year

Year	Postings	Employers	Duty lines (M)	Lines per posting	New (%)	New AI (%)
2019	192,069	9,421	2.04	10.6	4.8	0.06
2020	132,879	8,889	1.39	10.5	11.7	0.14
2021	216,577	11,250	2.33	10.7	12.8	0.19
2022	180,527	12,477	2.01	11.1	14.0	0.14
2023	142,375	12,059	1.62	11.4	14.7	0.18
2024	244,383	16,070	2.78	11.4	14.3	0.35
2025	289,967	17,969	3.35	11.6	14.7	0.80
2026[a]	253,983	15,673	3.11	12.3	17.1	1.70
All	1,652,760		18.62	11.3		

Notes: Postings in English with a description of at least 200 characters and at least one duty line, excluding bulk posters, one posting per multi-location advert. Employers: employer accounts (platform and employer identifier). New: duty lines with no close counterpart among the duty sentences of 2019H1 postings (Section 3.1); 2019 includes the reference half-year itself. New AI: new duty lines that describe AI work. [a]January to September.

2.4 Release

The release will contain, for each posting, its title, employer, dates, location, pay fields, occupation code with its confidence, seniority, industry, the crawl-intensity and representativeness weights, and flags for agencies, bulk posters, remote work and closed adverts; for each line of its text, its class and, for duty lines, the novelty score and AI labels; aggregates by quarter, occupation, state and platform; and the code for the whole build. It will not contain the text of the adverts, which is the employers' own writing: the text stays in Common Crawl's public archive, and each posting carries the address of its archived page (file, offset and length), from which the released code rebuilds it. The data will be released under CC BY 4.0 and the code under the MIT licence, with a later version of this paper.

3 Measuring new work

3.1 Duty lines with no counterpart in 2019

I embed every distinct duty sentence with a small sentence-embedding model (all-MiniLM-L6-v2; Reimers and Gurevych, 2019; Wang et al., 2020) and compute, for each, its highest cosine similarity to any of the 708,110 distinct duty sentences in postings first seen in the first half of 2019. A duty line is new if that similarity is below a threshold of 0.66, which makes 10% of the lines of the second half of 2019 new: the baseline churn of wording, as different employers describe the same or similar duties in different words. The threshold is a screen, not a judgement, and the lines it lets through are read (Section 3.3). To see what it means, Sonnet 5.5 read 1,000 duty lines of 2023–26 beside their most similar duty of early 2019, spread evenly over the range of similarity (Appendix Figure B1). Most pairs below the threshold are variants of an old duty; Sonnet reads 12–16% of the most distant pairs as new tasks, 4% of those at the threshold, and almost none above 0.75.

Weighting these readings by how many lines sit at each similarity, about 72% of the non-AI lines that Sonnet would call new tasks fall below the threshold, so the counts of new non-AI tasks below capture most but not all of them; the estimate rests on the one new task Sonnet found among 572 non-AI lines above the threshold. Thresholds that make 7% or 14% of 2019H2 new move the levels but not the comparisons (Appendix Table C1).

The share of new lines rises with distance from the reference, to about 14% by 2022, and it rises as much in the occupations that use AI least as in those that use it most. Some of this is genuine change, but most is variation in the objects, tools and wording of familiar duties: of the rise in new non-AI lines between 2019H2 and 2022 (4.0 points), Sonnet reads 8% as new tasks and the rest as variants, re-wordings and lines that are not duties (Section 4.3).

The measure finds new work where it is known to exist. Of the duty lines of 2020–21 that mention COVID-19, 64.5% are new. When the new lines of 2020–21, together with the COVID-19 policy notices that the text layer sets aside, are clustered into themes and the themes ranked by how many employers use them, 12 of the 14 that rise are about COVID-19: four are duties, from screening and safety protocols to vaccination programmes, and eight are notices, such as vaccination requirements (Appendix B). Most COVID-19 text in postings was not about duties: in 2021, 12% of sentences with a COVID-19 term were duty lines, and the rest were notices, legal text and statements about the company. COVID-19 duty lines peaked at 0.22% of all duty lines in early 2021.

3.2 AI work: building it or working with it

I find duty lines about AI in two steps. Keyword rules flag every duty sentence that mentions AI, machine learning, language models, generative AI or named AI tools. Claude Sonnet 5.5 then reads every one of the 134,656 flagged sentences, with the job title and occupation group but not the date, and records whether it describes AI work, which side of the technology it is on and what the worker does. Building AI covers developing, training and evaluating models and AI products; deploying and integrating them; and selling and supporting them. Working with AI covers using AI tools in one's own work; prompting, directing or reviewing AI systems and their output; and finding uses for AI, redesigning workflows around it and training colleagues in it. Governing AI (policy, risk and compliance) is a third side.

A second model (Claude Opus 5.5), reading 2,000 of the sentences blind, agrees with Sonnet on the side in 90% of cases. Keyword rules catch most work with AI tools from 2024 on but only about half of the work of building AI, which is often described without naming AI, so the building side is a lower bound (Appendix B.2). A classifier trained on all the labelled sentences extends the labels to every duty line.

3.3 Reading the new lines

Every label in this paper comes from a census or a designed random sample, and a second model re-reads random subsamples of the AI labels and of the readings of new lines (Appendix Table B3). For the new non-AI lines, Sonnet 5.5 reads a random 500 in every year from 2019H2 to 2026 and every third of occupations by AI use, enough to estimate a share of a few per cent to within about two percentage points, each beside the most similar duty of early 2019 and without the year or the cell, and judges whether it describes a new task, a variant of the old duty with a new object, tool or setting, the old duty in new words, or not a duty at all. Opus 5.5, reading 500 of the same

pairs blind, agrees with 86% of these judgements; it calls even fewer of the lines new tasks (0.8% against 2.0%) and rarely the same ones, so which lines are new tasks is the least certain part of the reading. The themes of the new lines give a second reading: I cluster the 416,000 distinct new sentences of 2025–26 in postings with confident occupation codes into 1,500 themes and keep those whose share of employers in 2025–26 is at least three times that of 2019–22. Sonnet names 90 of them, the 60 that rose most and the largest whose lines are mostly not about AI, and judges each to be AI work by side or, if not, a new task, a variant, a re-wording or not work.

I group occupations by observed AI use, the observed-exposure measure of Massenkoff and McCrory (2026): it counts the occupation's tasks that are rated feasible for language models by Eloundou et al. (2024) and that appear in conversations with Claude in 2025. It is therefore measured after the period of interest began, and it shares its task ratings with Eloundou et al.'s exposure measure. Thirds are weighted by postings, so each holds a third of them.

3.4 Duty lines and O*NET tasks

O*NET's task statements are the standard list of what each occupation does, and most measures of AI exposure are built on them. I map every duty sentence to its closest O*NET 27.1 task statement (2022) with the same embeddings, both among all 19,265 statements and among those of the posting's own occupation. A duty line has a related O*NET task when the closest task is similar enough that a language model is more likely than not to judge the two to describe at least related work: Sonnet 5.5 graded 1,530 such pairs spread over the range of similarity, and the threshold is where a logistic fit of its judgements crosses one half (Appendix Figure B7). The mapping gives an outside reference that does not move with the sample: a duty with no related O*NET task is new relative to the official description of work in 2022, not relative to early-2019 postings.

4 Results

4.1 How much new work

New work has grown where AI is used. Figure 1 splits the new duty lines of each year by what they are. In 2019H2, 9.9 in every 100 duty lines are new and not about AI, the baseline churn, and Sonnet reads almost all of them as variants, re-wordings or lines that are not duties. The share of new lines rises to about 14 in every 100 by 2022 and 17.8 in the second half of 2026, and most of the rise, even in 2026, is more of the same churn. Genuinely new work, the new non-AI tasks and the new AI lines, rose from 0.2 in every 100 duty lines in 2019H2 to 0.6 in 2022 and 2.3 in 2026 across all occupations (panel a). In the third of occupations with the most AI use it rose from 0.9 in 2022 to 2.6 in 2025 and 4.6 in 2026, of which 3.7 are AI work; in the third with the least use it stayed at or below 0.5 (panels b and c). Against the one recent shock that is known to have created new duties (Appendix Figure B2), three and a half years after ChatGPT new AI duty lines are about 9 times as common as COVID-19 duty lines were at their peak in early 2021, and still rising.[1]

A map of duty space shows the same pattern: between 2019–22 and 2025–26, holding the occupation

[1]Counting all text rather than duty lines, 16% of postings mentioned COVID-19 at the peak (2022H1), mostly in notices about vaccination and remote work, and 26% of postings mention AI in 2026; a COVID-19 term appeared in a duty line of at most 1.4% of postings, against 12% for an AI keyword in 2026.

mix fixed, one cluster of AI work gained more share of duty lines than any other part of the map (Appendix B.3). The ratio between the thirds does not depend on the sample's tilt towards computer occupations. With the occupation mix of employment the high-use level of new AI lines halves, to 1.8 in every 100 duty lines, but it remains about 13 times the low-use level (10–13 times across all the variants in Appendix Table C1).

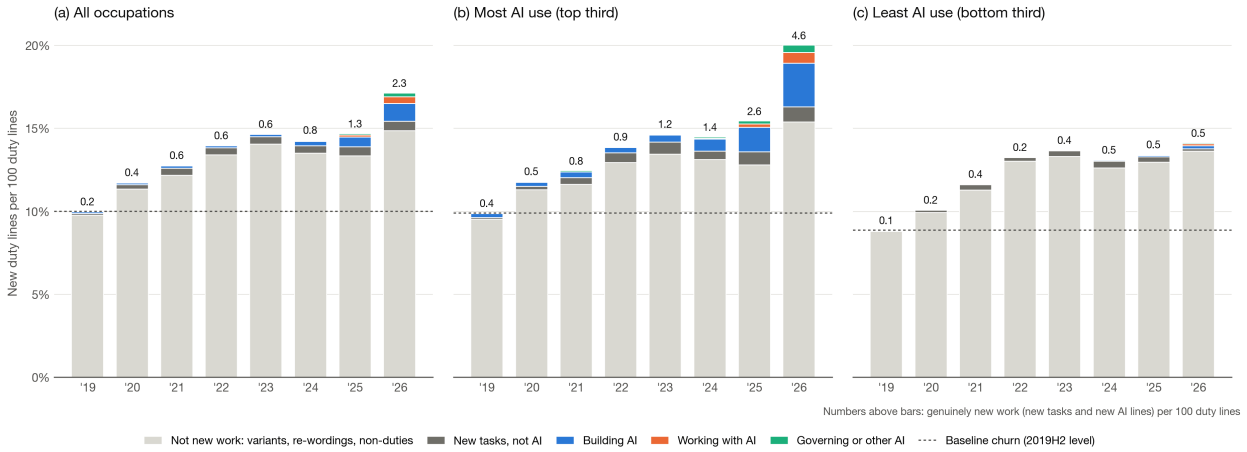


Figure 1: New work in duty statements. New duty lines (no close counterpart among the duty sentences of 2019H1 postings) per 100 duty lines, by year (2019: second half; 2026: January to September), split by what Sonnet 5.5 reads them to be: variants and re-wordings of an old duty and lines that are not duties, new tasks that are not about AI, and new AI lines by side. The split of the non-AI lines uses Sonnet's reading of 500 lines in every year and third of occupations by AI use. (a) All occupations. (b, c) The thirds of occupations with the most and least observed AI use. Dotted line: the panel's 2019H2 level, the baseline churn (10% in panel a, by construction). Numbers above the bars: genuinely new work (new tasks and new AI lines) per 100 duty lines.

4.2 Building AI or working with it

The new AI work is moving from the producers of AI to its users. Figure 2 shows the share of postings with at least one duty of each side, by occupation group. Building AI grew first and most in computer occupations: from about one in ten of their postings in 2019–22 to 33.8% in the second quarter of 2026. Working with AI barely existed before 2024. By mid-2026 it appears in 10.8% of postings in computer occupations and 9.2% in management and business occupations, where it is now as common as building AI, and in 1.8% of the rest.

What AI duties ask for has changed with it (Figure 3). AI work grew from 0.4 in every 100 duty lines in 2019 to 3.3 in 2026 (panel a). In 2019, building models and AI products made up 72% of AI duty lines; in 2026 it makes up 34%. The growth is in driving adoption, from 2% to 17%; in using AI tools, from 3% to 14%; in selling and supporting AI, from 6% to 12%; and in governing AI, from under 1% to 7% (panel b). Counting every duty line, working with AI was 5% of AI duty lines in 2022 and 24% in 2026, or 29–30% with employment or new-hire weights.

The balance differs by sector. In 2026, 26.9% of postings in information firms have a building-AI duty and 14.5% a working-with-AI duty, and in manufacturing building AI also leads (7.1% against 3.4%). In finance the two are level (8.8% and 8.5%), and in health care and retail working with AI is as common as building it (1.8% against 1.7%; 2.2% against 1.8%).

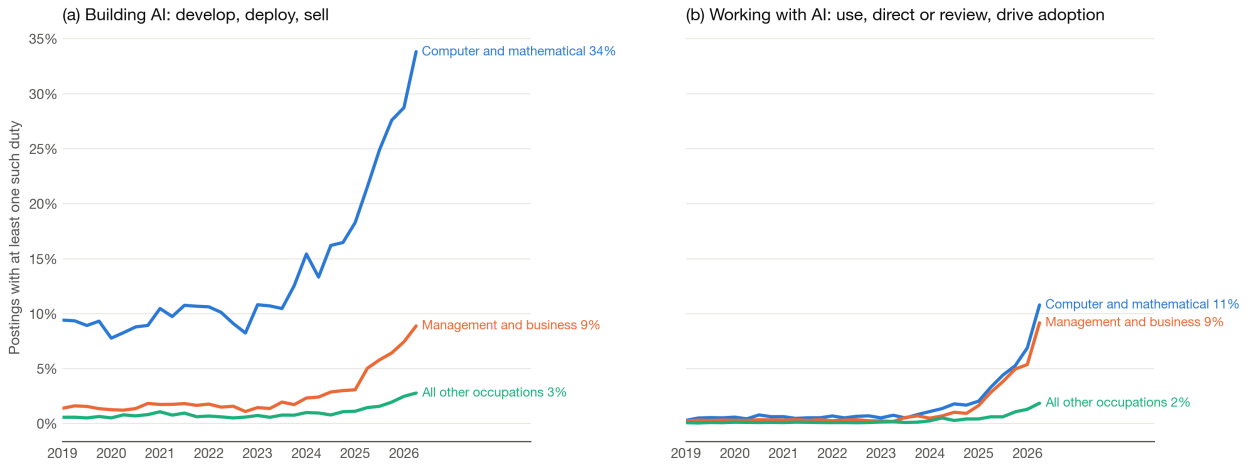


Figure 2: Share of postings with at least one duty line about building AI (a) or working with AI (b), by quarter and occupation group. Every duty line is classified: Sonnet 5.5 labels for AI-keyword sentences and a classifier trained on all labelled sentences for the rest. 2026Q3, a partial quarter, is omitted.

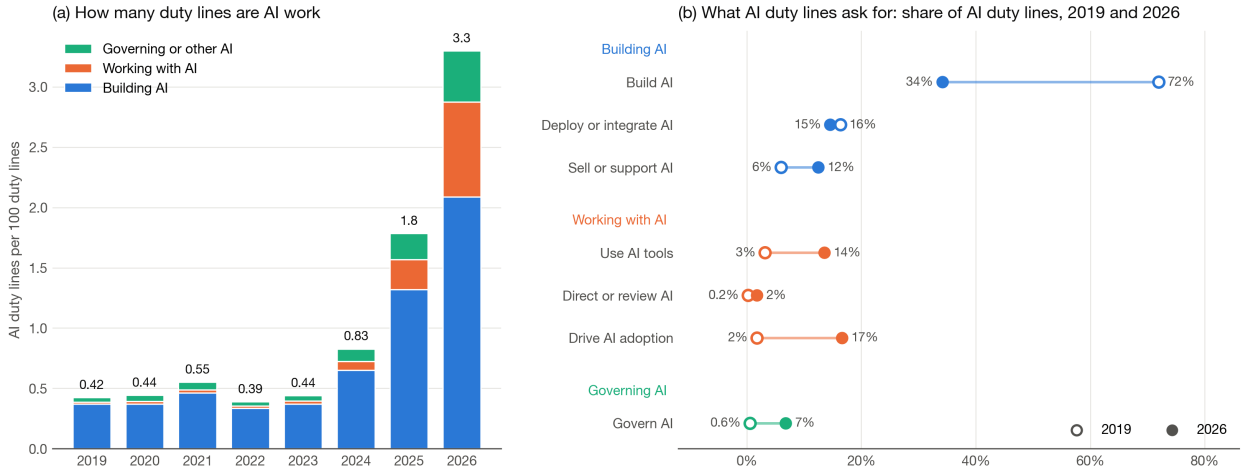


Figure 3: What AI duties ask for. (a) AI duty lines per 100 duty lines, by year and side; every duty line is classified (Sonnet 5.5 labels of sentences with an AI keyword, a classifier for the rest). (b) The activities of AI duty lines as shares of all AI duty lines, 2019 (open) and 2026 (filled): Sonnet 5.5 labels of every duty sentence with an AI keyword; the shares exclude AI work that fits no activity. Appendix Table C2 gives every year.

4.3 Little genuinely new work outside AI

Most of the new lines are not about AI, and most of those are not new work (Figure 4). In every year, Sonnet reads 67–72% of the new non-AI lines as variants of an old duty with a new object, tool or setting, 11–15% as the old duty in new words and 13–15% as not duties at all. New tasks are 1.2% of the new non-AI lines of 2019H2, when little new work is to be expected, and 3.6% of those of 2026. They rose in the occupations that use AI most, from 1.2% in 2019H2 to 5.6% in 2026, while in those that use it least they stayed at a few per cent. As a share of all duty lines in high-use occupations, new non-AI tasks were 0.6% in 2022 and 0.9% in 2026, a difference within sampling error. Most are technology work outside AI, such as blockchain and security tooling, and some are AI work that the AI labels missed, such as compressing models or building systems that

evaluate their output.

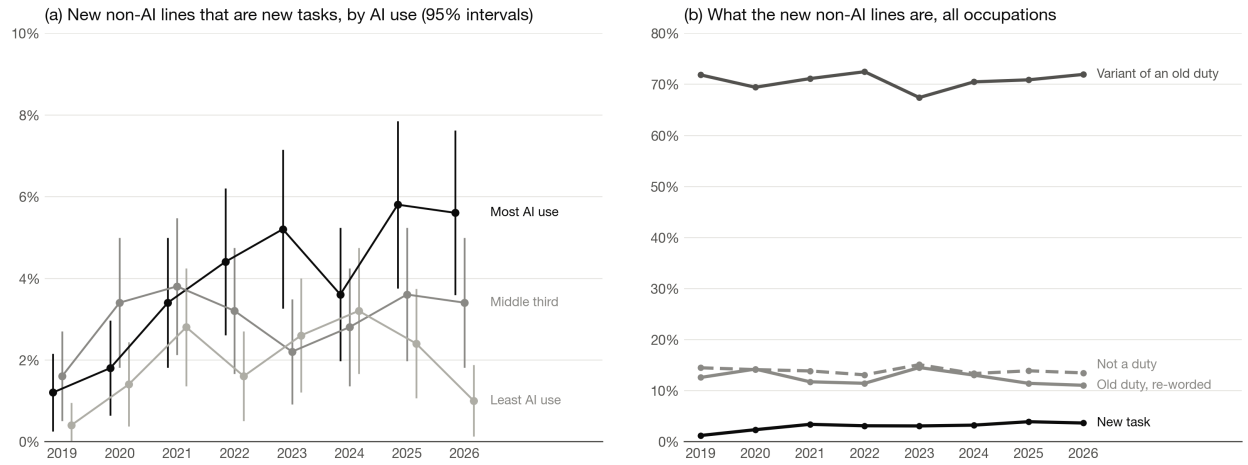


Figure 4: What the new non-AI duty lines are. Sonnet 5.5 compares each of 500 new non-AI lines in every year and third of occupations by AI use with the most similar duty sentence of 2019H1, without seeing the year or the cell. (a) Share of the lines read as new tasks, by third of observed AI use, with 95% intervals. (b) Share of each reading, all occupations (the thirds and the occupations without a measure of AI use, weighted by their new non-AI lines). 2019: second half.

The themes tell the same story (Figure 5). Of the 60 themes that rose most, 49 are AI work: integrating language models into applications, building AI platforms and agents, using AI tools to work faster, building AI-native workflows and championing AI adoption. Of the 40 named themes that Sonnet does not class as AI work, Sonnet judges 29 to be familiar work in new words or templates, such as codifying playbooks, owning a sales pipeline or shipping fast and iterating, 8 to be variants and 2 not to be work; 1 is a new task, automated security and compliance guardrails. Panel (b) shows the variants and the new task: they extend familiar engineering, security and go-to-market work to new tools and settings. The same method, run on 2020–21, finds COVID-19.

New job titles are where the new work concentrates (Figure 6). Five of the six largest titles that are new since 2022 or have grown fastest are AI titles, and they rose steeply from late 2024: forward deployed engineers were posted by 9 employers in 2019–22 and 247 in 2024–26, AI engineers by 10 and 168, and GTM engineers, a title absent from these postings before 2024, by 118 (Appendix Table C6). By the first half of 2026, 7.8 in every 1,000 employers posting were hiring a forward deployed engineer. The new titles also carry new duties. In the median new title, 74% of duty lines have no close counterpart among the duties of its three nearest established titles, against 33% for established titles (41% against their own 2019–22 postings); the gap is 16 points after controlling for the size of the comparison set.

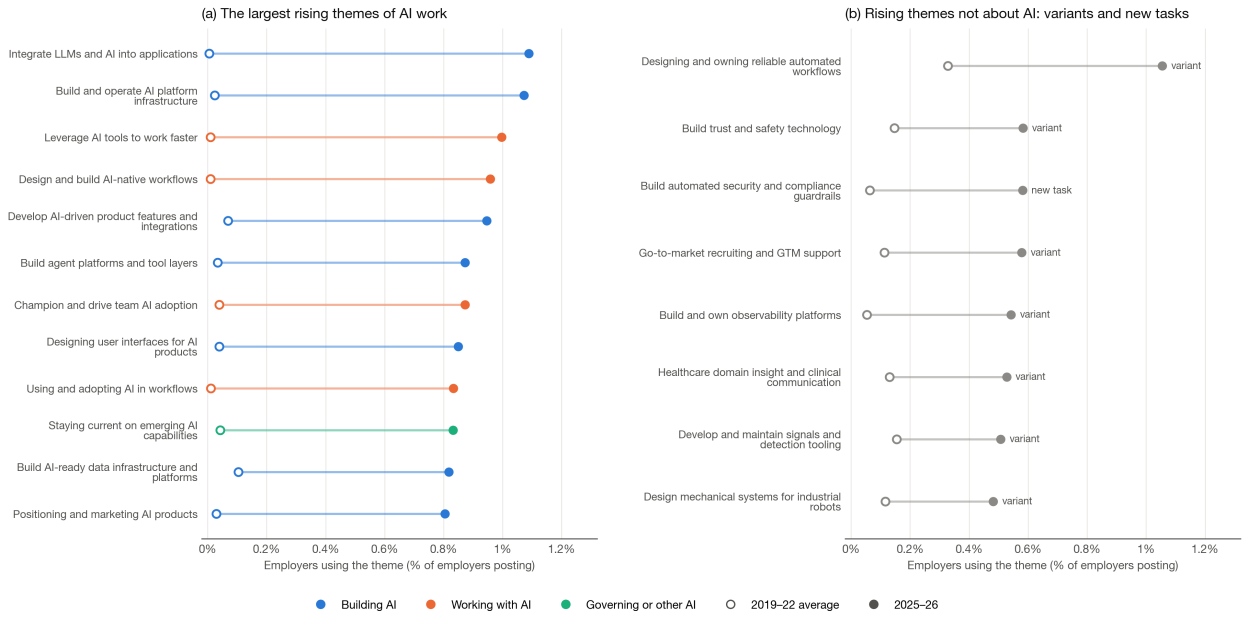


Figure 5: The largest rising themes of new duty lines: the share of employers using each theme in 2019–22 (open) and 2025–26 (filled). Themes: clusters of the distinct new sentences of 2025–26; rising: used by at least three times the share of employers. Names and judgements by Sonnet 5.5 from recent and older example sentences. (a) AI work, by side. (b) Themes not about AI that Sonnet judges variants of familiar work or new tasks; the re-worded themes are left out (Appendix Table C5).

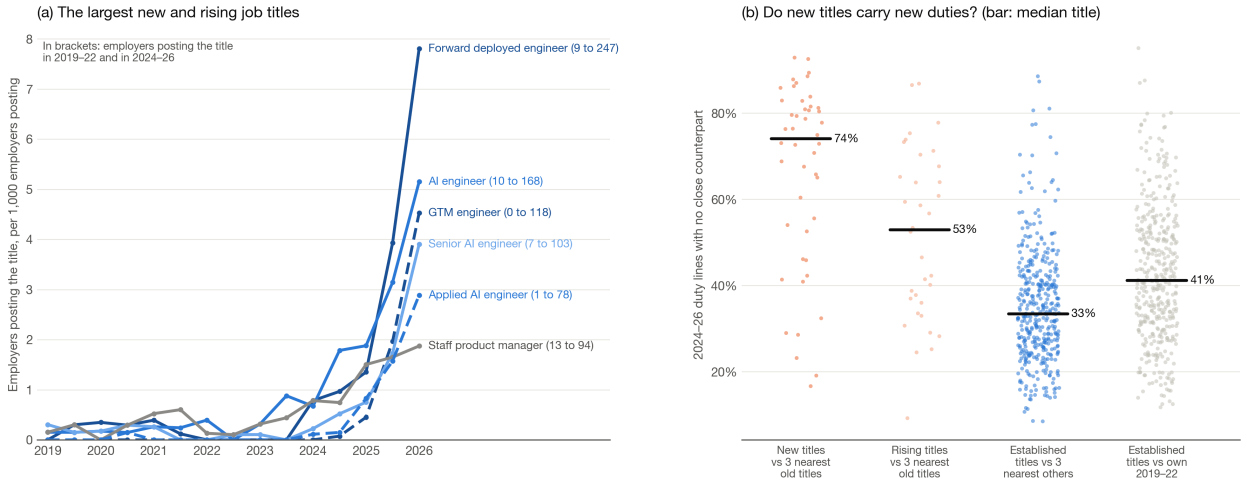


Figure 6: New job titles. (a) Employers posting the largest new and rising titles, per 1,000 employers posting in the half-year, 2019H1–2026H1; in brackets, employers posting the title in 2019–22 and in 2024–26. New titles: at least 10 employers in 2024–26 and none in 2019–22; rising titles: at least 20 employers in 2024–26 and five times their 2019–22 count. Blue: titles with an AI duty in at least half of their postings. (b) Share of 2024–26 duty lines with no close counterpart among the duties of the three most similar established titles (new and rising titles, and established titles against each other) or of the title's own 2019–22 postings; bar: median title.

4.4 Work that O*NET does not list

Measures of AI exposure take O*NET's list of tasks as given and ask which of them AI can do (Felten et al., 2021; Webb, 2020; Eloundou et al., 2024). The new AI duties mostly fall outside the list of the occupation that now does them (Figure 7). In its own occupation's O*NET list, 88% of continuing duty lines have a task that describes at least related work, against 64% of new lines not about AI and 50% of new building-AI lines and 54% of new working-with-AI lines. Across all occupations, most new AI lines clear the threshold for some task, but the closest tasks are often loose matches from other occupations: programming robots, building business-intelligence tools and developing machine-learning algorithms (Appendix Table B2). At the stricter standard of the same work, O*NET describes 12% of continuing duty lines and almost none of the new lines, about AI or not. The list has not drifted away from the rest of the work: among non-AI lines in the occupations that use AI most, the share with no related task in the occupation's list moved only from 17% in 2019 to 19% in 2026. So far, much of the new work of generative AI is work that the task lists behind exposure measures do not contain, and those measures miss where it reinstates labour.

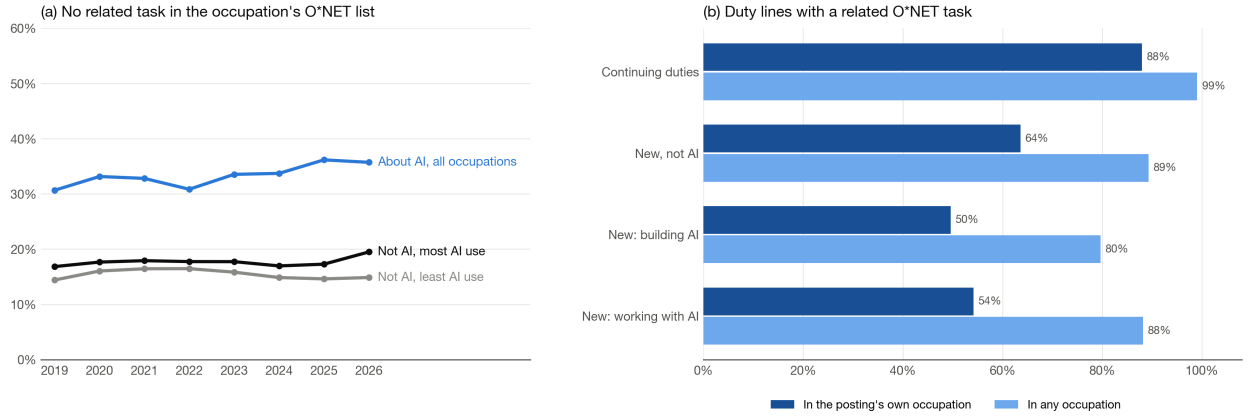


Figure 7: Duty lines and O*NET 27.1 task statements. A duty line has a related task when the closest task statement is similar enough that Sonnet 5.5 judges such pairs to describe at least related work more often than not (Appendix Figure B7). (a) Share of duty lines with no related task among the statements of the posting's own occupation, by year: lines not about AI in the thirds of occupations with the most and least observed AI use, and lines about AI in all occupations. (b) Share of duty lines with a related task in the posting's own occupation and in any occupation, by line class, 2019–26.

4.5 Where new work lands

Do occupations where AI is used more acquire more new work after generative AI arrives? I estimate, for 236 detailed occupations over half-years from 2019 to 2026,

$$y_{ot} = \alpha_o + \gamma_t + \sum_{t \neq 2022\text{H2}} \beta_t U_o + \varepsilon_{ot},$$

where U_o is the occupation's observed AI use, standardised, and y_{ot} an outcome in percentage points. Occupations are weighted by their 2019–22 postings and standard errors clustered by occupation. Figure 8 shows the coefficients. Working-with-AI duties are flat in 2019–22 and rise from late 2024: by the second half of 2026 (to September), an occupation one standard deviation higher in AI use

has 3.3 points more postings with such a duty. Averaged over 2024–26 against 2019–22 the effect is 1.2 points (standard error 0.4), or 0.8 (0.3) with major-group-by-period effects. New building-AI lines rise from 2023 and new non-AI lines only in 2026.

The design is exploratory. The junior and senior shares and posting volumes also differ by AI use, but those differences began before 2022 (Appendix Figure C2). And it cannot say whether new work follows the use of AI or the capability of the technology: when observed use and the exposure measure of Eloundou et al. (2024), which rates what a model could do, enter together, working-with-AI postings load more on exposure (0.96, standard error 0.33) than on use (0.56, standard error 0.52) in the 228 occupations with both measures. The two correlate at 0.64 across the 738 occupations with both measures, and since observed use only counts tasks that Eloundou et al. rate as feasible, the comparison cannot separate them.

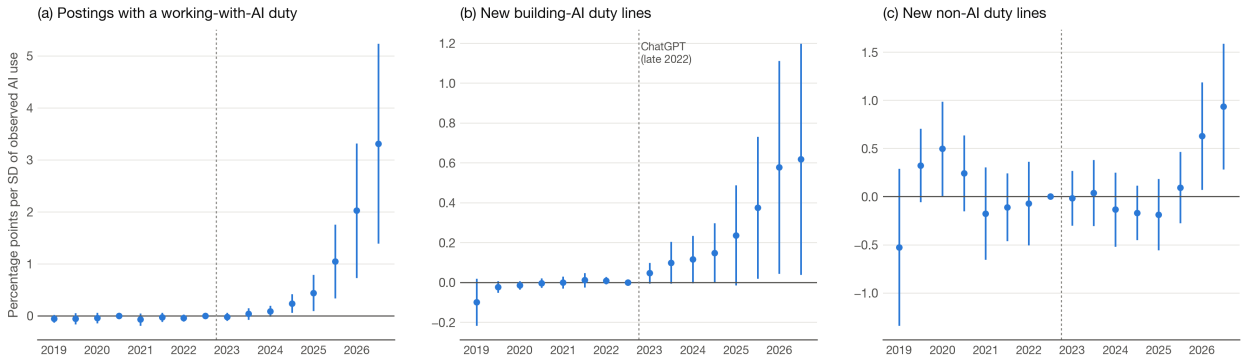


Figure 8: New work by observed AI use of the occupation: coefficients β_t on standardised AI use (Anthropic Economic Index) by half-year, relative to 2022H2, with 95% intervals. 236 occupations with confident codes, 3,657 occupation-half-years. (a) Postings with a working-with-AI duty. (b) New building-AI duty lines. (c) New non-AI duty lines.

4.6 Who gains

The new work is better paid, mostly because of who does it. Figure 9 reports the posted-pay premium of postings with a building-AI or a working-with-AI duty, on 436,705 postings that state pay, as the comparison tightens. Against postings in the same occupation, state and year, building-AI postings advertise 0.21 log points more and working-with-AI postings 0.08. Within the same firm, occupation and year the premia fall to 0.08 and 0.03, and within the same title at the same firm to 0.03 and 0.03. Working with AI carries no premium in computer occupations, but 0.11 log points within firm and occupation in occupations outside computing and management (Table C4).

The postings are also becoming more senior: the share of junior titles fell from 24% of postings in 2022 to 20%, and the share of senior and staff titles rose from 14% to 19%. The shift began in 2024. It was larger in occupations that use AI more, but those occupations were already becoming more senior before 2022 (Appendix Figure C2), so the occupation design cannot attribute it to AI. Inside firms the evidence is weaker still: after a firm's first posting for AI-building work, neither the working-with-AI duties of its non-technical postings nor the junior share of its other postings changes detectably, and the timing of such first postings is tied to bursts of captured postings, so I report the firm design in Appendix C.1.

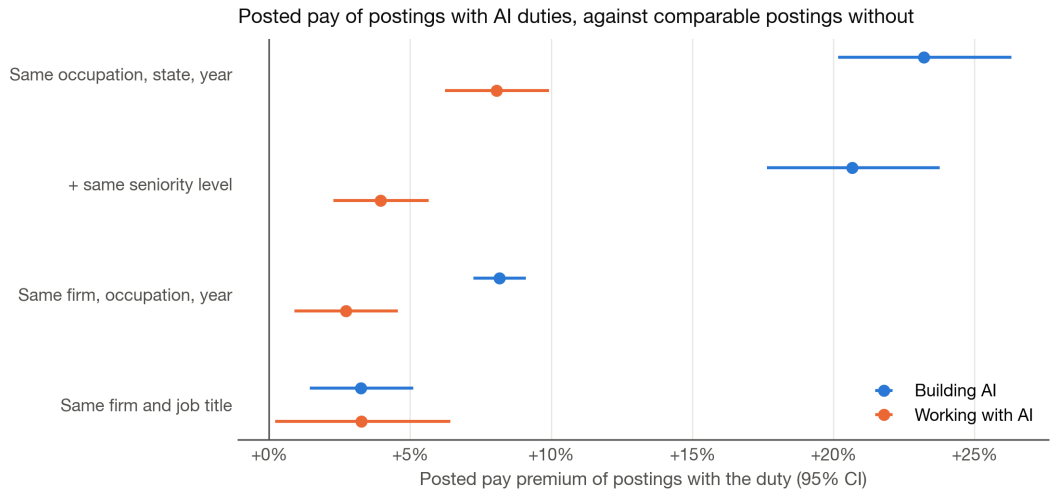


Figure 9: Posted-pay premium of postings with an AI duty, 2023–26, as the comparison tightens: log annual posted pay on indicators for a building-AI and a working-with-AI duty, with fixed effects for the cells named on the left; premia in per cent, $100(e^{\beta}-1)$, with 95% intervals (the text reports log points). Standard errors clustered by firm.

5 Limitations

The postings describe what employers ask for when they hire, not the work that people do, and only for the employers that use these platforms. They over-represent professional, technical and large employers, and their platform mix shifted: Workday postings carry text only from 2022, and Lever's pages stop late in 2025. The facts survive dropping both platforms, holding the platform mix fixed and reweighting to the occupation mix of employment or hiring, but the levels depend on the weights (Appendix Table C1).

Every label in the paper comes from a language model or from a classifier trained on its labels. A second model agrees with most of the first's labels and finds even fewer new non-AI tasks, though rarely the same ones, and the labels reproduce known facts such as the COVID-19 duties; keyword-free AI work is under-counted on the building side.

Novelty is measured against one reference period, so the share of new lines grows with time; I address this by having a language model read the lines rather than counting them. The last half-year is partial, and several results sharpen in 2026: without it, the occupation-level effect on working-with-AI duties falls from 1.2 to 0.8 points. Finally, the occupation-level design is descriptive: observed AI use is measured in 2025, it is built on the same task ratings as AI exposure, and occupations that use AI more differ in other ways.

6 Conclusion

Whether a new technology raises the demand for labour depends on the new work it creates (Acemoglu and Restrepo, 2018, 2019), and in the past most new work appeared where technologies complemented what workers already did (Lin, 2011; Autor et al., 2024). For generative AI, the evidence has so far come from what the technology could do and what people use it for. Read directly in job postings, its new work is concentrated in the occupations that use it and is mostly

about AI itself: first building it, now increasingly using it and driving its adoption, and in health care, finance and retail working with AI is already as common as building it. Little of the new work is of other kinds.

Two implications follow. Measures that score O*NET's tasks for what AI can do (Felten et al., 2021; Webb, 2020; Eloundou et al., 2024) see the tasks it may take over better than the ones it creates, since half of the new AI duties have no related task in the list of the occupation that now performs them; following reinstatement requires reading what employers ask for. And whether working with AI becomes a durable part of many jobs, or a passing stage before further automation, depends on how fast the technology's capabilities grow (Kwa et al., 2025). The derived data behind these facts will be released, so that this question can be followed as the technology diffuses.

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A The dataset

This appendix summarises the construction and quality of the postings. The build covers January 2017 to September 2026; this paper uses postings first seen from 2019. Fuller documentation, including the extraction rules and the occupation classifier's training labels (posting identifiers and codes), will accompany the release.

A.1 Construction

Table A1 lists the platforms and Table A2 accounts for every captured page. Table A3 gives counts and field completeness by year. Counts follow Common Crawl's crawling, not the labour market: from September 2017 to January 2019 the archive captured Workday and iCIMS pages at several times their later rate, iCIMS pages before September 2017 carry no text, and 2023 had fewer crawls than the years around it. I therefore start the analysis in 2019 and do not interpret posting counts as vacancies.

Table A1: Platforms

Platform	US ATS pages (%)	US postings	Years	Description (%)	Salary (%)	State (%)	Verdict
iCIMS	36.2	1,393,482	2017–2026	88	17	84	yes
Workday	20.2	634,446	2017–2026	26	11	66	caveats
Greenhouse	10.3	446,489	2017–2026	100	44	83	yes
Jobvite	4.1	171,110	2017–2026	100	21	85	yes
JazzHR	3.2	96,274	2017–2026	84	30	78	caveats
Breezy HR	2.6	99,834	2017–2026	95	30	87	yes
Lever	2.5	117,637	2017–2026	100	31	71	caveats
UKG (UltiPro)	2.3	69,272	2017–2026	76	16	74	caveats
SmartRecruiters	2.2	83,018	2017–2026	78	12	83	caveats
Dayforce	1.9	76,437	2019–2026	97	23	93	yes
Paylocity	1.3	56,490	2017–2026	100	33	93	yes
Ashby	1.2	51,563	2020–2026	79	50	55	caveats
Paycom	1.0	38,143	2019–2026	93	37	86	caveats
Recruitee	0.3	7,677	2018–2026	94	36	91	caveats

Notes: US ATS pages: the platform's share of the US job pages captured on the 20 platforms first queried, in one crawl per quarter from 2017 to 2026. US postings: postings in the release. Description, salary, state: share of the platform's US postings with a description of at least 200 characters, salary information and a resolvable state.

Table A2: From captured pages to postings, 2017–2026

Step	All	US	Non-US	Unresolved
Captured job pages on the 14 platforms	10,505,800			
not fetched or not parsed (kept)	3			
Jobs (platform, employer and job ID)	5,205,625	3,630,815	1,348,062	226,748
repost merge (same title, location and text)	−350,263	−285,346	−63,262	−1,655
Postings	4,855,362	3,345,469	1,284,800	225,093
non-US postings dropped	−1,284,800		−1,284,800	
Postings kept (US and unresolved)	3,570,562	3,345,469		225,093
iCIMS portal merge	−4,163	−3,597		−566
Release (version 1.1)	3,566,399	3,341,872		224,527
of which: closed-ad notice (flagged)	34,995			
of which: no title	24,232			
of which: no description of 200+ characters	888,023			

Notes: Counts at each step of the build, which covers 2017–2026. Jobs are called US, non-US or unresolved before the repost merge, postings after it.

Table A3: US postings by first-seen year

		Weighted		Field present (%)						Median
Year	Postings	postings	Employers	Description	Duties	Date	Employer	State	Salary	length
2019	274,531	225,538	10,739	87	69	80	75	91	6	3,142
2020	197,526	197,526	10,311	85	64	84	75	88	7	3,217
2021	308,332	308,332	12,408	88	70	87	84	88	13	3,456
2022	287,957	287,957	13,161	87	69	85	78	81	25	3,594
2023	192,502	192,502	12,606	91	70	88	82	83	35	3,691
2024	326,187	326,187	16,624	90	68	87	84	82	42	4,045
2025	371,142	371,142	18,314	88	69	85	79	78	48	4,551
2026	327,003	327,003	16,065	87	69	84	79	75	51	4,967

Notes: Weighted postings use the crawl-intensity weight, which scales crawls that captured a platform at an unusual rate to the platform's 2019–20 capture rate. Description: at least 200 characters. Duties: a duties section was found by header (the text layer of Section 2 replaces this rule). 2026: January to September.

A.2 Coverage and representativeness

Figure A1 summarises the benchmarks. Where and when states required pay ranges in postings, the share of postings with pay information jumped (panel a); the share offering remote or hybrid work rose with the pandemic, as in Lightcast's and Indeed's series, if to a higher level (panel b); posting shares by detailed occupation track Lightcast's (panel c); and median posted pay tracks the occupation's median wage (panel d). Figure A2 compares the postings with hires, Figure A3 with the occupation mix of employment and of other posting data, and Table A4 summarises the differences as dissimilarity indices: the share of postings that would have to change category for two distributions to match. Figure A4 compares states, and Table A5 the shares of postings that state pay or offer remote work against Indeed's and Lightcast's series. Live postings per crawl fell by 39% between February and May 2020 and track the Indeed postings index until 2023 but not after (Figure A5), as the share of each platform that the archive captured changed. Table A6 compares extracted fields with Web Data Commons' independent parse of the same pages.

Table A4: Dissimilarity indices

Comparison	Level	ATS postings weighted by		
		crawl intensity	OEWS	hires
ATS vs OEWS employment	major group	0.34	0.00	0.14
	detailed	0.45	0.32	0.35
ATS vs Lightcast postings	major group	0.22	0.21	0.28
	detailed	0.38	0.38	0.42
Lightcast vs OEWS	major group	0.21		
	detailed	0.35		
ATS vs CES employment, 2024	state	0.12		
ATS vs JOLTS openings, 2024	state	0.13		

Notes: Share of postings that would have to change category for the two distributions to match. Occupations: postings first seen in 2023–26; OEWS employment in May 2025. Columns: crawl-intensity weight, and the representativeness weights that give each major group its share of OEWS employment or of new hires in the CPS (the OEWS weights set the major-group index against OEWS to zero by construction). States: 2024.

Table A5: Fit of the pay-information and remote-work series

Series	Period	Quarters	Correlation			Gap (points)		
			levels	changes	changes[a]	2021	2023	2025
Pay information vs Indeed	2020–26	27	0.98	0.24	0.30	−11.9	−6.0	−1.7
postings with a description	2020–26	27	0.98	0.19	0.32	−10.0	−2.1	5.0
Remote work vs Lightcast	2019–26	31	0.88	0.14	0.33	3.3	1.8	7.7
Remote work vs Indeed	2019–26	31	0.85	0.48	0.33	3.8	3.2	9.5

Notes: Quarterly shares of postings against Indeed's pay-transparency and remote-work trackers and Lightcast's remote-work share. Changes: first differences. [a]Postings dated by their date posted where available.

Table A6: Agreement with Web Data Commons

Field	Route	Matched pairs	Compared	Agree (%)
title	URL (Workday)	2,628	7	
	page template	7,577	7,543	99.4
	structured data	102,617	102,579	100.0
	all	112,822	110,129	99.9
US call	metadata or description	1,954	3	
	page template	9,918	7,486	100.0
	structured data	100,950	100,855	100.0
	all	112,822	108,344	100.0
location	URL (Workday)	2,628	1,104	99.7
	page template	7,577	7,497	97.4
	structured data	102,617	97,036	97.2
	all	112,822	105,637	97.2
description	page template	8,489	6,525	63.8
	structured data	104,333	104,321	100.0
	all	112,822	110,846	97.8

Notes: Web Data Commons postings on the fourteen platforms in the October 2022 and October 2024 crawls, matched by platform, employer and job identifier. Route: where the field came from in this build.

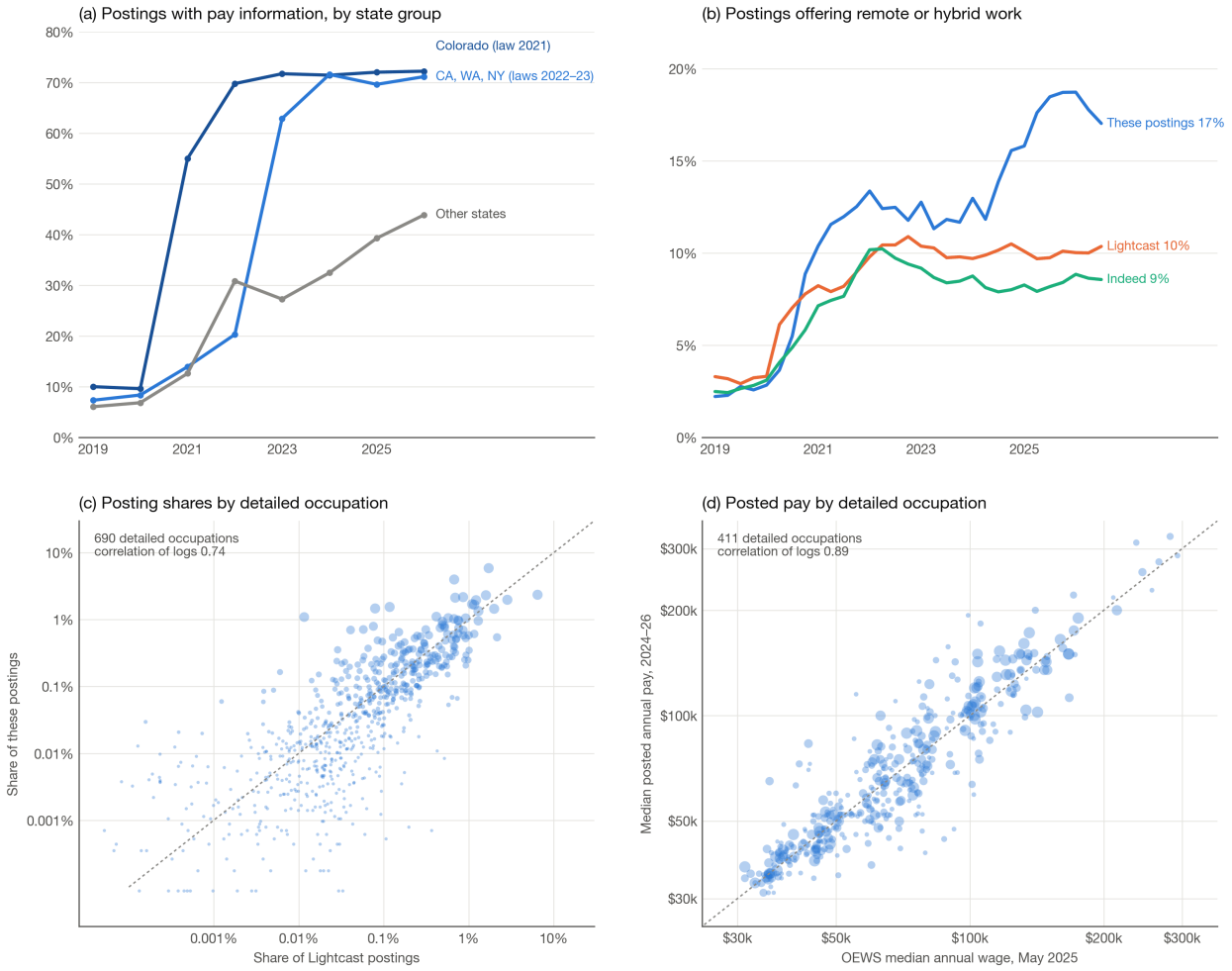


Figure A1: The postings against known facts. (a) Share of postings with pay information by state group; Colorado's pay-transparency law took effect in 2021, California's, Washington's and New York's in 2022–23. (b) Share of postings offering remote or hybrid work, by quarter, against Lightcast's and Indeed's series. (c) Shares of postings by detailed occupation against Lightcast's, 2023–26 (log scales). (d) Median posted annual pay, 2024–26, against the OEWS median wage by detailed occupation (log scales). Dotted lines: equality.

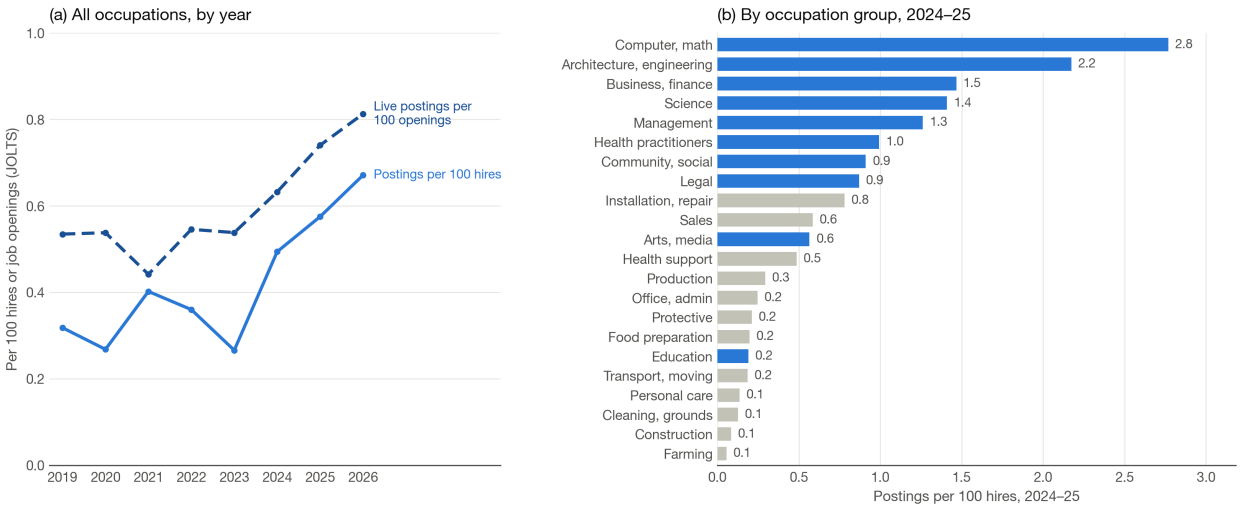


Figure A2: Coverage of US hiring. (a) Postings per 100 hires in JOLTS, and live postings per crawl per 100 job openings. (b) Postings per 100 hires by occupation group, 2024–25; blue: professional and technical groups.

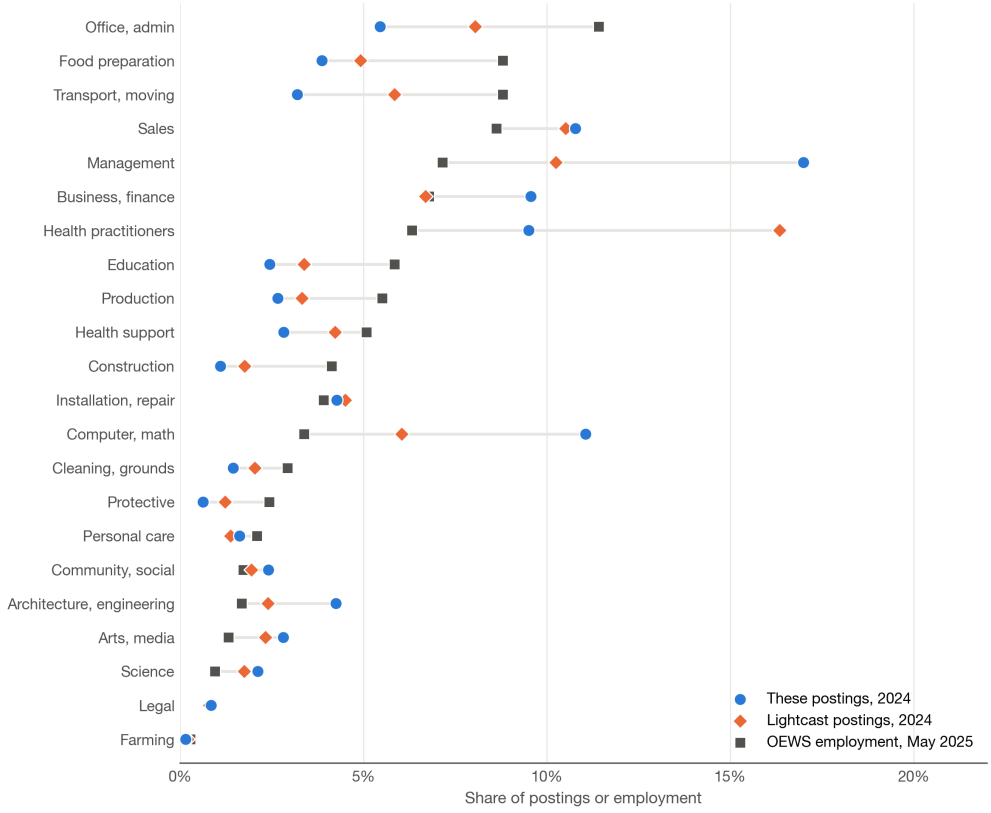


Figure A3: Occupation mix by SOC major group: these postings and Lightcast postings in 2024, and OEWS employment in May 2025.

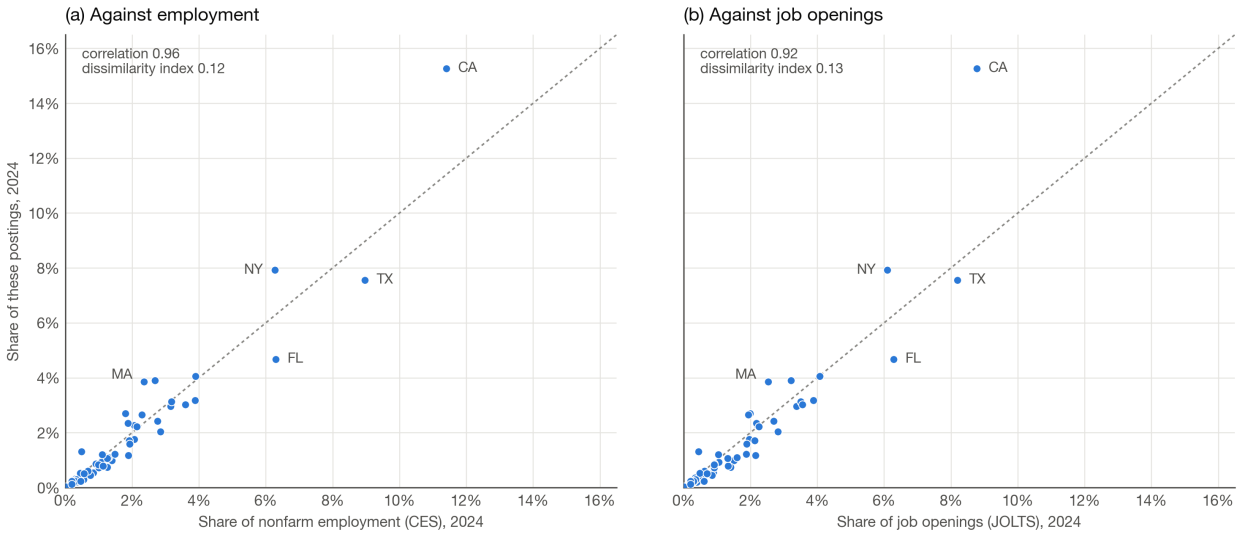


Figure A4: State shares of postings against state shares of nonfarm employment (a) and of job openings (b), 2024. Dotted lines: equality.

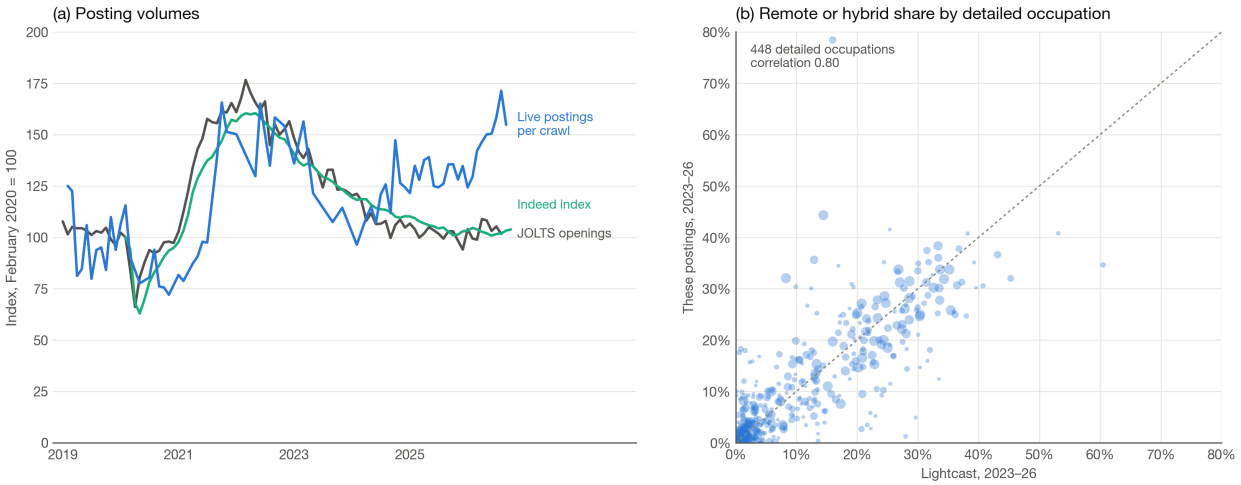


Figure A5: (a) Live US postings per crawl, JOLTS job openings and the Indeed postings index (February 2020 = 100). (b) Share of postings offering remote or hybrid work by detailed occupation, against Lightcast, 2023–26. Dotted line: equality.

A.3 Occupation codes

The classifier uses the title's words and character sequences and the first 400 characters of the description. Its training data are 30,001 postings coded by Sonnet 5, stratified by platform and year, and 30,000 postings whose titles match an O*NET title. Figure A6 shows that its confidence is calibrated and how agreement rises with the confidence threshold. Most disagreements are between neighbouring codes.

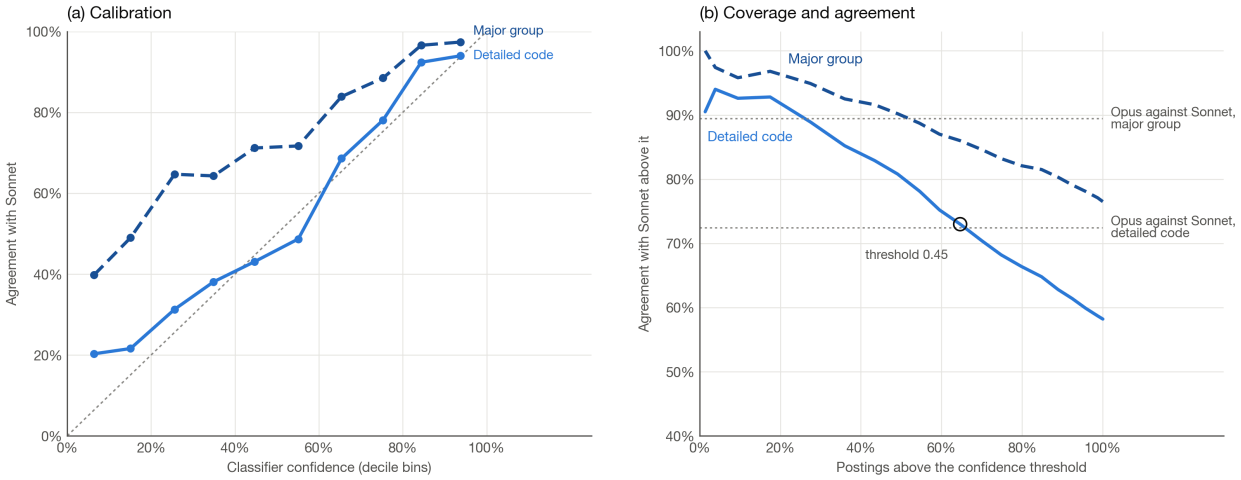


Figure A6: (a) Agreement with the language model by classifier confidence decile. (b) Share of test postings above each confidence threshold and agreement on them; dotted lines: agreement between two language models.

A.4 The text layer

Table A7 reports the duty classifier's agreement with the language-model labels by year. The labels cover 1,000 postings drawn by year and 200 with flat text; a retest of 100 postings agrees on 99% of duty labels. The weakest class is statements about the role, a fifth of which the classifier calls duties.

Table A7: Duty classifier against language-model labels

Year	Postings	Units	Precision	Recall	Duty share, Sonnet (%)	Predicted (%)
2017	100	3,067	0.90	0.94	32.0	33.3
2018	100	3,359	0.92	0.95	30.9	31.9
2019	100	3,525	0.92	0.93	28.3	28.7
2020	100	3,669	0.92	0.95	30.9	32.0
2021	100	3,351	0.94	0.93	27.9	27.8
2022	100	3,927	0.92	0.92	20.0	20.0
2023	100	3,799	0.94	0.94	24.1	24.0
2024	100	3,694	0.92	0.96	26.7	27.7
2025	100	4,379	0.91	0.96	23.7	25.1
2026	100	4,695	0.89	0.94	23.2	24.3
All (weighted)	1000	37,465	0.92	0.94	26.4	27.1
Flat text	283	10,148	0.92	0.93	21.3	21.5

Notes: Five-fold cross-validation grouped by posting. Units: sentences and headers. Precision: share of predicted duties that the language model labels duties; recall: share of the language model's duties predicted. Duty share: share of units labelled or predicted duty.

A.5 Known gaps

Workday pages carry descriptions only where they include structured data, almost none before 2022 and about half from 2023, and their titles and locations come from the page address. Common Crawl holds almost no UKG pages in 2021 and almost no Lever or Paycom postings in 2026. Four tested platforms whose archived pages mostly carry no job data, Taleo, ADP, Workable and BambooHR, are not included, nor are ATS platforms outside the 20 first queried, and school districts, public employers and health systems mostly post elsewhere. About 1% of occupation codes are SOC 2010 rather than 2018 codes.

B Measurement

B.1 Novelty and the COVID-19 check

The embedding model is all-MiniLM-L6-v2 (mean pooling, 128 tokens, unit-normalised vectors). The search is exact: every distinct duty sentence of 2019H2–2026 is compared with every distinct duty sentence of 2019H1. Figure B1 shows what the similarity means. Sonnet 5.5 read 100 duty lines of 2023–26 in each bin of 0.05 of similarity to their most similar duty of 2019H1, from 0.45 to 0.95, beside that duty, with the prompt of the new-line reading (Appendix B.5). Re-wordings dominate above 0.75, variants between 0.55 and 0.75, and new tasks and lines that are not duties are most common in the most distant pairs. Among non-AI lines above the threshold, Sonnet calls 1 of 572 a new task; below it, 4%.

Figure B3 shows the COVID-19 check. Sentences with a COVID-19 term rose in 2020 and peaked in 2022, but most were policy notices, legal text and statements about the company; duty lines with a COVID-19 term peaked at about 20 in every 10,000 duty lines in 2021 (panel a). The themes of the new lines of 2020–21, which include the policy notices, show both kinds (panel b): four rising themes are COVID-19 duties and eight are notices. Figure B2 sets the new AI lines of 2023–26 against the COVID-19 duty lines of 2020–22.

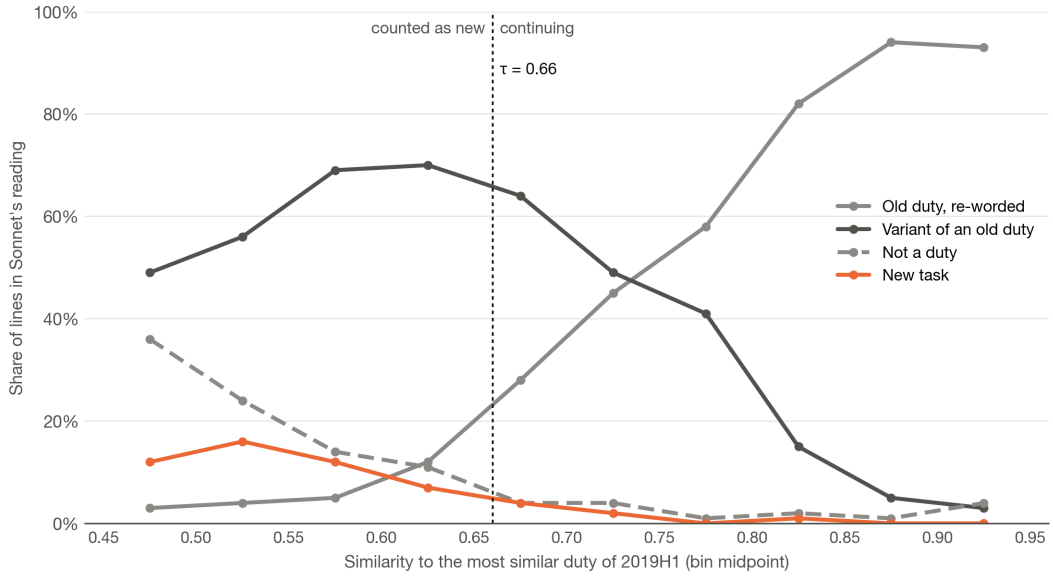


Figure B1: What similarity to 2019 means: Sonnet 5.5's reading of 1,000 duty lines of 2023–26, each beside its most similar duty of 2019H1, 100 per bin of similarity. Dotted line: the novelty threshold.

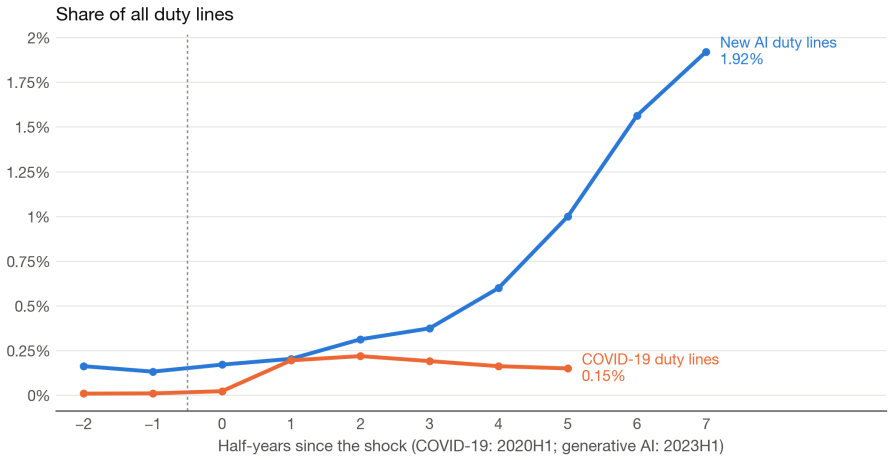


Figure B2: New AI duty lines and COVID-19 duty lines as a share of all duty lines, in half-years since each shock (COVID-19: 2020H1; generative AI: 2023H1). COVID-19 duty lines: duty lines with a COVID-19 term; notices of COVID-19 policy, such as vaccination requirements, are not duty lines and are not counted. COVID-19 series to 2022H2.

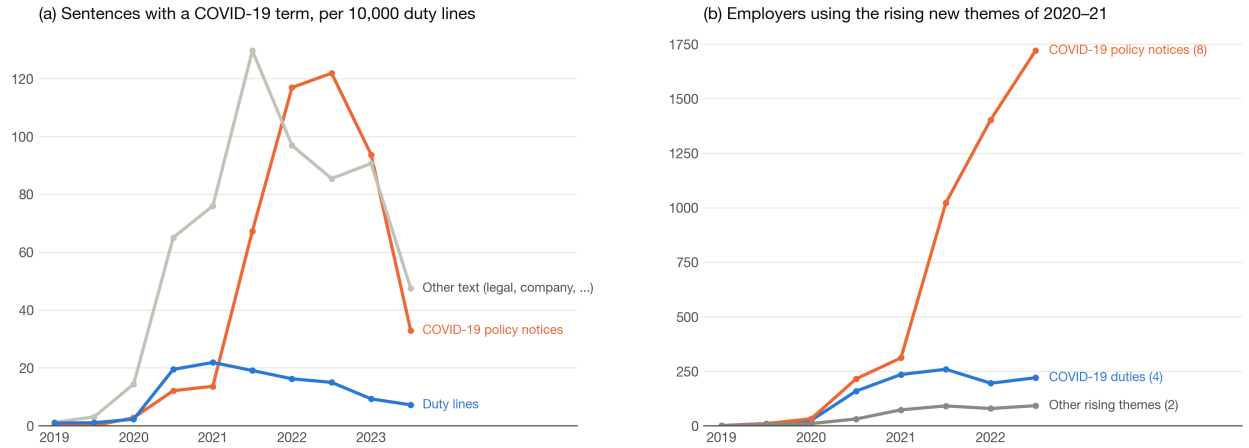


Figure B3: The novelty measure finds COVID-19. (a) Sentences with a COVID-19 term per 10,000 duty lines, by the text layer's class: duty lines, COVID-19 policy notices and other text. (b) Employers using the rising new themes of 2020–21, by whether Sonnet 5.5 judged the theme to be COVID-19 duties, COVID-19 notices or neither. Themes: clusters of the new lines and notices of 2020–21, rising when used by at least three times as many employers as in 2019H2.

B.2 AI labels

On 2,000 AI sentences drawn at random and rated blind by both models, Opus 5.5 and Sonnet 5.5 agree on whether the sentence describes AI work in 98% of cases, on the side in 90% (Cohen's κ 0.82) and on the activity in 85%, and they give the same aggregate shares. AI work without an AI keyword is not rare: Sonnet finds it in 0.6% of a random sample of 139,346 duty sentences without a keyword (18,750 lines a year) and in 8.5% of sentences with near-miss words such as "agent", "automation" or "model". Table B1 scales the AI rate of the random sample to all keyword-free lines: the keywords catch 35–52% of building-AI work, depending on the year, but 86–87% of working-with-AI work in 2025–26. The classifier that extends the labels to every duty line agrees with Sonnet on the labelled sentences (precision 0.98, recall 0.95) but finds few of the AI sentences without keywords (recall 0.16), so the building side remains a lower bound.

Table B1: Share of AI work caught by the keyword rules

Year	Building AI	Working with AI	Any AI work	Keyword AI lines
2019	38 [33, 45]	40 [18, 66]	41 [35, 47]	4,690
2020	35 [30, 41]	72 [31, 94]	40 [35, 46]	3,452
2021	39 [34, 45]	57 [26, 83]	44 [39, 50]	7,155
2022	51 [42, 59]	68 [27, 92]	52 [44, 60]	4,403
2023	43 [36, 49]	59 [28, 84]	45 [38, 51]	4,143
2024	49 [44, 54]	79 [57, 92]	54 [49, 59]	16,353
2025	52 [48, 56]	86 [75, 93]	59 [55, 63]	44,211
2026	51 [48, 54]	87 [81, 91]	61 [59, 64]	78,784

Notes: Percent of AI duty lines that contain an AI keyword, with 95% intervals from the binomial interval of the random sample's AI rate. Keyword AI lines: duty lines with an AI keyword that Sonnet labels AI work. Random sample: 18,750 keyword-free duty lines per year.

B.3 A map of duty space

To show where duties sit and where they change, I draw 40,000 duty lines a year from 2019 to 2026 at random, embed their 287,234 distinct sentences with the same model, and place them in two dimensions with UMAP (cosine distance, 30 neighbours; McInnes et al., 2018). A map of all 10.2 million distinct sentences would be impractical to fit and unreadable as a scatter, and drawing the same number of lines each year gives every year equal weight. Nearby points describe similar work; the axes have no units. Figure B4 names 30 regions of the map (k-means clusters of the map, named by Sonnet 5.5 from random sentences in each, without dates). Figure B5 shows where the share of sampled duty lines grew and shrank between 2019–22 and 2025–26, with the occupation mix held fixed: each line of 2025–26 is weighted by its SOC major group's share of the 2019–22 lines over its share of the 2025–26 lines, so the map shows changes in what occupations ask for rather than in which occupations post. The labelled clusters are among 300 finer clusters, with changes of at least four standard errors and at least 15 employers on the side that is larger. One cluster of AI work gained 1.2 percentage points of the sampled lines, more than any other part of the map; recruiting and nursing-documentation duties lost most (0.22 and 0.21 points), while patient and family care coordination gained (0.16). Figure B6 places the AI duties of 2026 on the map: most sit at its top edge, next to data and engineering work, and some in sales, marketing, administrative and patient-care regions.

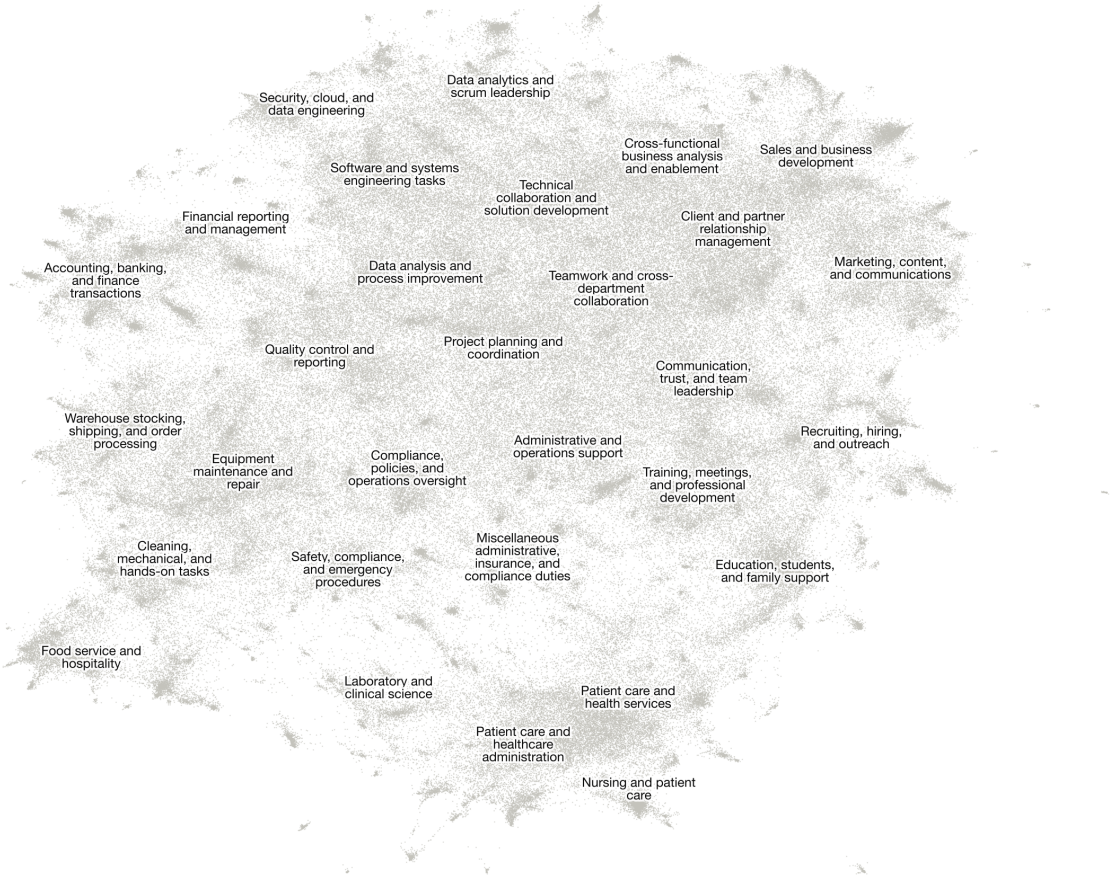


Figure B4: A map of duty space: 287,234 distinct duty sentences from a random sample of 320,000 duty lines (40,000 a year, 2019–2026), placed by meaning (sentence embeddings, UMAP). Region names by Sonnet 5.5 from random sentences in each of 30 regions.

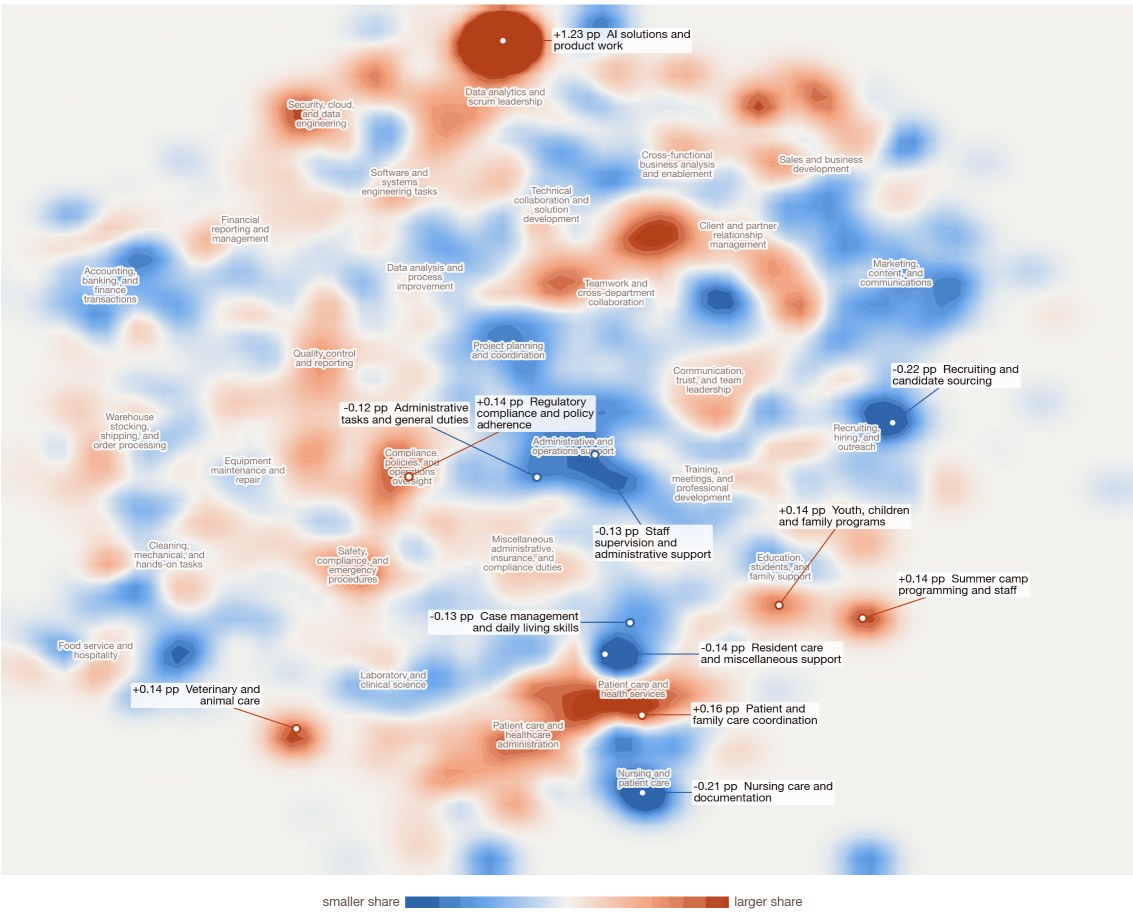


Figure B5: Where duties grew and shrank between 2019–22 and 2025–26, with the occupation mix held fixed: smoothed change in the share of sampled duty lines across the map of Figure B4. Labelled: the largest gains and losses among 300 finer clusters, with changes of at least four standard errors and at least 15 employers, in percentage points of all sampled lines; names by Sonnet 5.5.

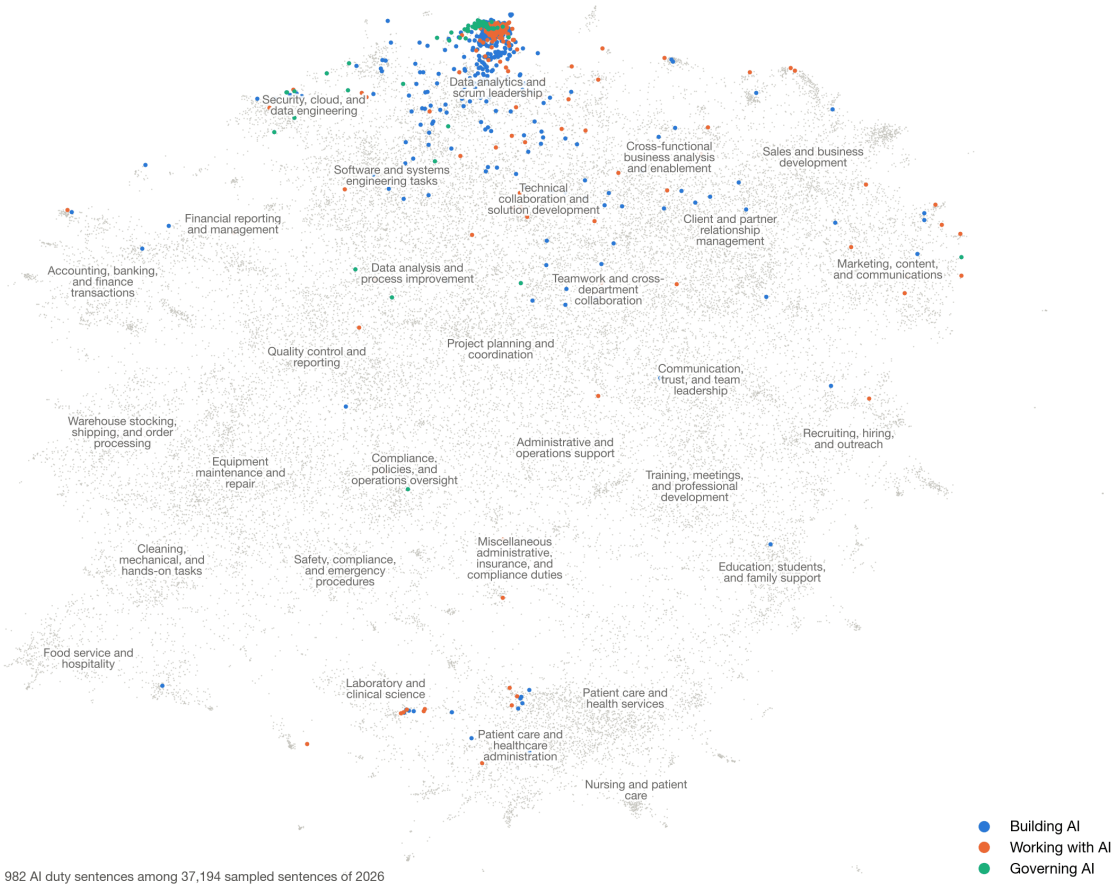


Figure B6: AI duties of 2026 on the map of Figure B4, by side (Sonnet 5.5 labels of sentences with an AI keyword). Grey: all sampled sentences of 2026.

B.4 Duty lines and O*NET tasks

Figure B7 shows how the matching thresholds were set. Sonnet 5.5 graded 85 duty lines in each bin of 0.05 of similarity to their closest O*NET task, separately for the closest task among all occupations and among the posting's own occupation, with the prompt of Appendix B.5 (2: the same work; 1: related work; 0: unrelated). The thresholds where a logistic fit of the share graded at least related work crosses one half are 0.44 among all tasks (95% bootstrap interval 0.42–0.46) and 0.35 within the occupation (0.32–0.37). The share graded the same work crosses one half only at 0.73 and 0.71, at the top of the graded range, so these two thresholds extrapolate the fits; at that standard, O*NET describes 12% of continuing duty lines and almost none of the new lines. Version 1 of this paper used thresholds of 0.53 and 0.44 from grades by Sonnet 5 of 440 duty lines of an earlier text layer; at those thresholds, 21% of new building-AI lines and 24% of new working-with-AI lines have a related task among all occupations, against 89% of continuing lines. Table B2 lists the O*NET task statements that the new duty lines of 2025–26 in high-use occupations are most often matched to, by the number of employers.

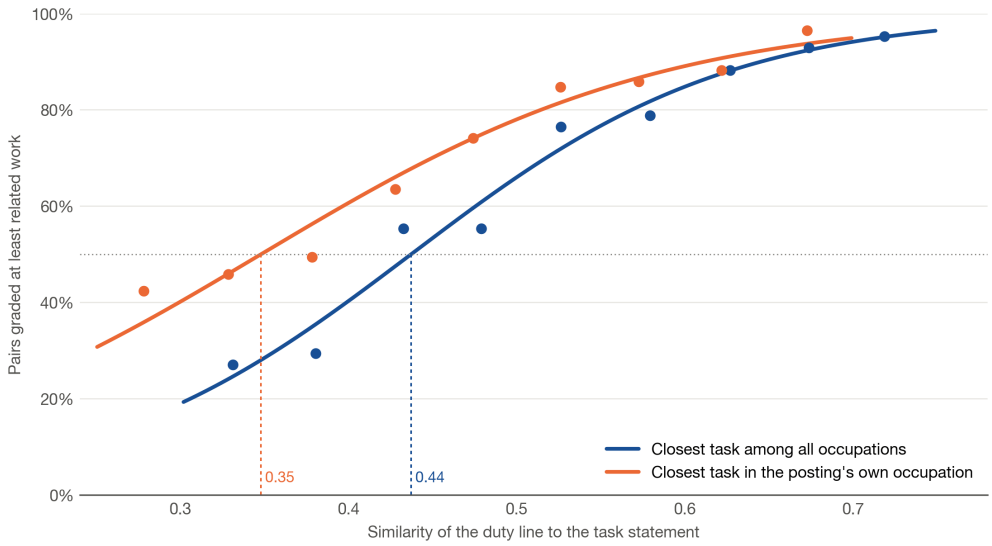


Figure B7: Setting the O*NET matching thresholds: share of duty-line and task pairs that Sonnet 5.5 grades at least related work, by similarity, with logistic fits, for the closest task among all occupations and in the posting's own occupation; 85 pairs per bin. Dotted lines: the thresholds, where the fits cross one half.

Table B2: O*NET tasks closest to new duty lines, 2025–26, occupations with high AI use

Closest O*NET 27.1 task	Employers	Lines
A. New AI duty lines		
Develop, test, or program new robots.	737	1,450
Train others to install, use, or maintain robots.	431	720
Create back-ups of robot programs or parameters.	421	719
Create business intelligence tools or systems, including design of related datab...	390	550
Collaborate with system architects, software architects, design analysts, and ot...	328	505
Collaborate with product development teams to research, model, validate, or impl...	279	454
Help clients improve decision making, abstract reasoning, memory, sequencing, co...	246	367
Develop or apply data mining and machine learning algorithms.	246	385
Develop defense plans or tactics, using intelligence and other information.	230	390
Research computerized automotive applications, such as telemetrics, intelligent ...	212	295
B. New duty lines not about AI		
Develop or implement marketing strategies.	569	766
Provide expertise to help financial institutions manage risks and maximize retur...	554	1,082
Collaborate with vendors to obtain or develop desired products.	534	944
Provide customers or installation teams guidelines for implementing secure syste...	497	972
Provide customers with ongoing technical support.	490	819
Collaborate with product development teams to research, model, validate, or impl...	477	780
Serve as liaisons between organizations, shareholders, and outside organizations...	463	650
Implement security practices to preserve assets, minimize liabilities, or ensure...	391	584
Provide training and support to system designers and users.	380	568
Develop or track quality metrics.	362	484

Notes: New duty lines (no close counterpart in 2019H1 postings) whose closest O*NET 27.1 task statement, among all occupations, describes at least related work by the threshold of Figure B7; ranked by distinct employers.

B.5 Language-model labels

Table B3 lists every labelling task. All labels come from Claude Sonnet 5.5, with Opus 5.5 as the second rater, except the occupation codes, which come from Claude Sonnet 5. The models are called through Anthropic's command-line client at medium reasoning effort, with structured output and every response cached, and they never see the year of the posting being labelled. The system prompts follow; line breaks are added for printing.

Table B3: Language-model labelling tasks

Task	Unit	Design	Labelled	Model	Second rating
Side and activity of AI work	Duty sentence with an AI keyword	Census	134,656	Sonnet 5.5	Opus 5.5, random 2,000: side 90%
AI work without a keyword	Duty line without an AI keyword	Random, 18,750 lines a year	139,346	Sonnet 5.5	
AI work, near-miss words	Duty line with a near-miss word	Random, 5,000 lines a year	35,166	Sonnet 5.5	
What new non-AI lines are	New non-AI line and nearest 2019H1 duty	Stratified: year × third of AI use, 500 a cell[a]	14,000	Sonnet 5.5	Opus 5.5, random 500: 86%
Novelty threshold	Duty line and nearest 2019H1 duty	Stratified: 100 per 0.05 of similarity	1,000	Sonnet 5.5	
O*NET matching thresholds	Duty line and closest O*NET task	Stratified: 85 per 0.05 of similarity, two searches	1,530	Sonnet 5.5	
Theme names and kinds	Rising theme of new lines	The 60 that rose most and the 40 largest not about AI	90	Sonnet 5.5	
Region names	Region or cluster of the map	Census of the 30 regions and the labelled clusters	51	Sonnet 5.5	
Seniority	Job title	Half in proportion to postings, half uniform	30,000	Sonnet 5.5	Title rules: 87%
Industry	Firm	Census of firms with at least three postings	30,539	Sonnet 5.5	Second pass on low or medium confidence
Duty lines (text layer)[b]	Every sentence of a posting	100 postings a year and 200 with flat text	1,200	Sonnet 5.5	Retest of 100: 99%
Occupation codes[b]	Posting	Stratified by platform and year	30,001	Sonnet 5	Opus, 500: detailed 72%

Notes: Labelled: items labelled (sentences, lines, pairs, themes, titles, firms or postings). [a]Eight years (2019: second half) by the three thirds of observed AI use, and 250 a year in occupations without a measure. [b]From the dataset (Section 2).

AI work: side and activity. Each call sends 60 items as "id | job title | occupation group | sentence".

```
You code duty sentences from US job postings for research on how AI changes work. Each item gives a
  job title, an occupation group and one duty sentence from that posting. Code every item.

ai: true if the sentence describes work that involves AI or machine-learning systems: generative AI,
  LLMs, chatbots, copilots, AI agents, machine-learning or deep-learning models, computer vision,
  speech or NLP systems, AI-driven automation. false for automation without AI (scripts, RPA,
  macros, workflow rules), statistics or analytics without ML, "models" meaning financial, data,
  business or 3D models, "agents" meaning people (insurance, customer service, real estate), and
  "AI" in other senses.

activity (if ai is true, the main one; if ai is false, "none"):
  build: research, design, develop, train, fine-tune or evaluate AI models, agents or AI-powered
    products and features that the organisation builds
  deploy: integrate AI into products, systems or infrastructure; MLOps; data pipelines for AI;
    operate, maintain or monitor AI systems in production
  sell_support: sell, market, consult on, implement for clients, or support AI products or services
  govern: AI policy, risk, compliance, ethics, security, legal or responsible-AI oversight
  use_tool: use AI tools to do the person's own work (drafting, coding with an assistant, analysis,
    research, design, answering customers)
  direct_review: direct, prompt or orchestrate AI agents or tools, or review, verify, correct or
    approve AI-generated output, as part of the person's own work
  adopt_enable: find AI use cases, lead or drive adoption of AI tools in a team or function,
    redesign workflows around AI, or train and coach colleagues to use AI
  other_ai: anything else involving AI (e.g. "stay current on AI trends")

gen (if ai is true): "generative" if it clearly involves generative AI (LLMs, chatbots, copilots,
  generative agents, text, image or code generation); "other_ml" if it is clearly other machine
  learning; "unclear" otherwise. If ai is false: "na".
```

New lines and the novelty threshold. Each call sends 40 items as "id | job title | recent sentence | most similar 2019 sentence".

```
You compare a duty sentence from a recent US job posting with the most similar duty sentence found
  in US job postings from early 2019, to judge whether the recent sentence describes new work. For
  each item give:
kind:
  new_task: the recent sentence describes an activity, responsibility or object of work that the
    2019 sentence does not, and that would not have been common in 2019 job postings for this kind
    of job
  variant: the same kind of activity as the 2019 sentence, with a different object, tool, setting or
    level of detail that is not new in itself
  reworded: the same activity as the 2019 sentence, in different words
  not_a_duty: the recent sentence is not a description of work (company, benefits, role context,
    requirements, or a fragment)
source (only for new_task; otherwise "na"): what makes it new: technology (non-AI tools, platforms,
  data systems), regulation_compliance, health_safety, product_market (new products, services,
  markets or business models), organisation_people (e.g. diversity, remote or hybrid work,
  wellbeing, culture), sustainability, other
```

O*NET matching thresholds. Each call sends 20 duties, each with its job title and one candidate task statement.

```
You judge how well standard task descriptions from the O*NET database describe a duty from a US job
  posting. For each candidate, give a grade:
2 = it describes the same work as the duty (wording may differ; small differences in scope are fine)
1 = related work (same area or a closely connected step), but not the same work
0 = unrelated, or only superficially similar
Judge the work itself, using the job title for context. Grade every candidate independently.
```

Themes. Each call sends up to eight recent and four older example sentences per theme.

```
You classify recurring duties in US job postings. Each group is a set of similar duty sentences from
  different postings: "recent" sentences, and the "older" sentences closest to the same theme. For
  each group give a short name (at most 8 words, sentence case, describing the recent work) and one
  category for the recent sentences:
ai_building: building, deploying, integrating, selling or supporting AI systems (models, agents, LLM
  products)
ai_using: using, prompting, directing, reviewing or driving the adoption of AI tools in the worker's
  own work
ai_other: governing AI, or other work about AI
new_task: genuinely new work that is not about AI and would not have been common in 2019 postings
  for such jobs
variant: a familiar kind of work with a new object, tool, platform or setting that is not new in
  itself
reworded: familiar work in new wording, or a shared template
not_work: culture, values, benefits, notices or generic statements rather than work
source (only for new_task; otherwise "na"): technology (non-AI), regulation_compliance,
  health_safety, product_market, organisation_people, sustainability, other
```

B.6 Seniority, industry and firms

Seniority levels are intern, junior, mid, senior, staff or principal, manager, director and executive. Title rules alone agree with Sonnet on 87% of 1,000 titles; the title model, trained on 30,000 Sonnet-labelled titles (half drawn in proportion to postings, half drawn uniformly from the remaining titles), agrees on 88% in cross-validation. Industry codes were checked and recoded where Sonnet's

confidence was low or medium, using the employer's company-description sentences; 77% of firms (90% of postings) are coded with high confidence. Employer accounts are linked across platforms when a normalised name is unique to one account on each of at least two platforms.

C Further results

Table C1: Robustness of the headline numbers

	New AI, high	New AI, low	Δ new non-AI	Working with AI (%)	Postings (%)	Exposure design
Baseline	3.71	0.35	+2.8	23.8	5.5	1.23 (0.39)
OEWS employment weights	1.79	0.14	+1.4	28.6	2.5	1.23 (0.39)
CPS new-hire weights	1.65	0.12	+1.4	30.0	2.5	1.23 (0.39)
Without Workday and Lever	3.71	0.35	+3.2	24.0	5.7	1.25 (0.38)
Fixed platform mix	2.87	0.21	+1.4	25.1	4.2	1.18 (0.39)
Novelty threshold −0.02 (7% of 2019H2 new)	2.78	0.26	+2.0	23.8	5.5	
Novelty threshold +0.02 (14% of 2019H2 new)	4.59	0.43	+3.6	23.8	5.5	
Without 2026H2	3.40	0.32	+2.2	21.6	4.6	0.81 (0.27)

Notes: New AI, high and low: new AI duty lines per 100 duty lines in 2026, thirds of occupations with the most and least observed AI use. Δ new non-AI: change in new non-AI lines per 100 duty lines, high-use occupations, 2022 to 2026 (raw, not read). Working with AI: share of AI duty lines, 2026. Postings: share of postings with a working-with-AI duty, 2026. Exposure design: post (2024H1 on) minus pre coefficient for postings with a working-with-AI duty, per standard deviation of AI use (standard error); the weights by year and major group do not change it, by construction. The novelty threshold does not affect the working-with-AI share or the postings column, which count all AI lines.

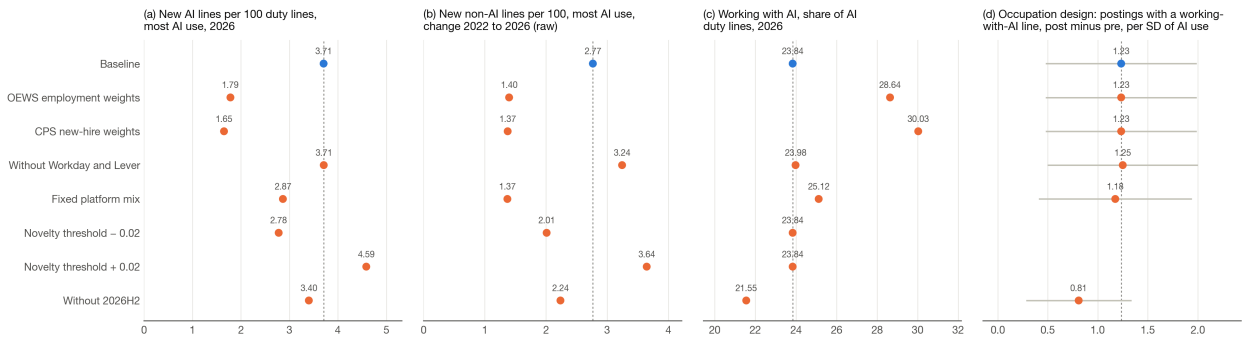


Figure C1: The first four headline numbers of Table C1 under each variant. Blue: baseline.

Table C2: The activities of AI duty lines by year (%)

Year	Build	Deploy	Sell, support	Use tools	Direct, review	Drive adoption	Govern	Lines
2019	71.9	16.3	6.0	3.2	0.2	1.8	0.6	4,162
2020	71.9	15.8	5.1	4.1	0.5	2.0	0.6	2,905
2021	64.8	21.8	6.9	3.3	0.4	1.8	1.1	5,954
2022	63.3	20.7	7.7	3.3	0.3	2.2	2.6	3,968
2023	63.0	18.2	9.4	3.4	0.6	2.8	2.6	3,679
2024	56.2	16.2	11.6	4.4	2.8	4.4	4.3	14,613
2025	47.2	17.6	11.6	7.9	1.0	9.9	4.9	39,847
2026	34.2	14.6	12.5	13.6	1.8	16.6	6.8	71,844

Notes: Sonnet 5.5 labels of every duty sentence with an AI keyword, as shares of the AI duty lines with an activity. Lines: AI duty lines with an activity. 2026: January to September.

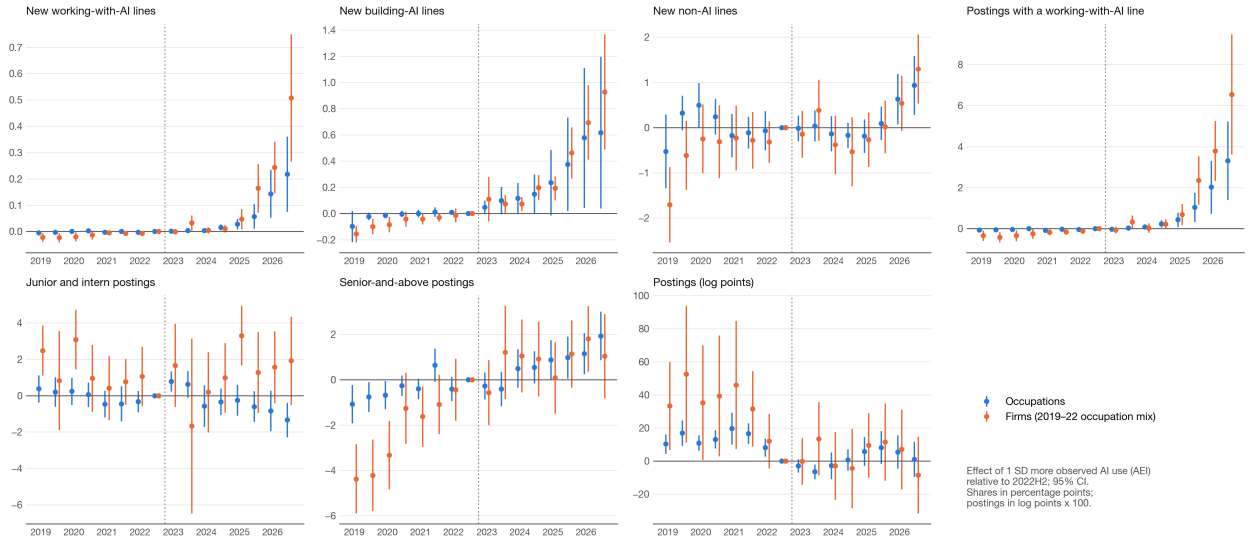


Figure C2: Occupation-level design for every outcome, and a firm-level version in which a firm's AI use is that of its 2019–22 occupation mix (236 occupations; 1,826 firms). Senior and above includes managers, directors and executives.

C.1 Firm event studies

If working with AI spreads within firms, it should appear in a firm's other jobs once the firm starts building AI. I compare firms whose first posting for AI-building work falls in 2023–25 with firms that start later, before those firms' own first such posting, and do the same around the first posting for working with AI, an event close to the adoption measure of Hosseini and Lichtinger (2026). Version 1 of this paper used a stacked design with later adopters as controls (Cengiz et al., 2019). Here I re-estimate it on a balanced panel of firms and quarters, with quarters without postings filled in, with the estimators of Callaway and Sant'Anna (2021), with not-yet-treated firms as controls and cohorts observed for six quarters after the event, and of Sun and Abraham (2021), with firms that start in 2026 as the comparison cohort. Figure C3 and Table C3 show the results for 380 firms that first post AI-building work in 2023–25 (535 firms in all) and 538 that first post working-with-AI work (957); the balanced Callaway–Sant'Anna estimates use the cohorts of 2023Q1–2024Q2, which are observed for six quarters after their event.

Neither robust estimator finds a detectable change. Working-with-AI duties in the firms' non-technical postings move by +0.4 points with Callaway and Sant'Anna's estimator and +0.9 with Sun and Abraham's (standard errors 0.4 and 0.5), against 2.0 in the stacked design of version 1, with no difference before the event. The junior share of their other postings moves by −0.4 and +0.1 points (standard errors 1.7 and 1.7) on a base of 17.5%. Around the first working-with-AI posting, the junior share moves by −2.0 points with Sun and Abraham's estimator (standard error 1.1), while Callaway and Sant'Anna's balanced estimate is +1.4 and fails its pre-trend test; Hosseini and Lichtinger (2026) measure junior employment from résumés, and these data the composition of postings. The design has a limit that no estimator removes: a firm's first such posting is more likely to be seen in a quarter when the archive captures many of its postings, so the event coincides with a burst of postings (59 log points in the event quarter) and posting volumes trend before it.

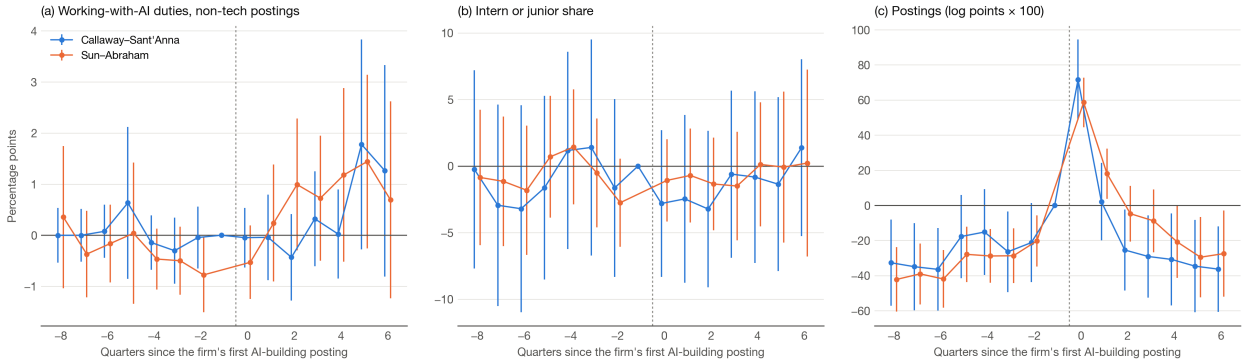


Figure C3: Firm event study around the first posting for AI-building work: coefficients by quarter relative to the event, with 95% intervals. Blue: Callaway and Sant'Anna's estimator with not-yet-treated controls, universal base period and cohorts observed for quarters 0 to 6. Orange: Sun and Abraham's estimator with firms that start in 2026 as the comparison cohort. Balanced panel of firms by quarter, 2019–2025, standard errors clustered by firm. Outcomes: (a) the share of non-technical postings without an AI-building duty that have a working-with-AI duty; among postings without AI duties, (b) the intern or junior share and (c) postings (log points × 100). Table C3 also reports the stacked design and the first working-with-AI posting.

Table C3: Firm event studies: post minus pre

Outcome	Stacked	Callaway–Sant'Anna	Pre-trend p	Sun–Abraham	Pre-trend p
A. Around the first AI-building posting					
Working with AI, non-tech postings	2.0 (0.5)	0.4 (0.4)	0.66	0.9 (0.5)	0.35
Intern or junior share	−0.4 (1.2)	−0.4 (1.7)	0.92	0.1 (1.7)	0.47
Senior or staff share	0.2 (1.4)	3.3 (1.7)	0.80	0.5 (1.6)	0.42
Computer occupations share	1.1 (1.1)	1.0 (1.8)	0.62	2.4 (1.7)	0.99
Postings (log points)	14.5 (6.1)	14.6 (7.9)	0.08	30.6 (7.5)	0.00
B. Around the first working-with-AI posting					
Intern or junior share	−1.5 (0.9)	1.4 (1.4)	0.01	−2.0 (1.1)	0.77
Senior or staff share	0.2 (1.1)	−1.0 (1.7)	0.21	−0.1 (1.3)	0.55
Computer occupations share	1.2 (0.9)	2.1 (1.4)	0.00	0.1 (1.3)	0.86
Postings (log points)	10.0 (4.6)	2.4 (9.1)	0.00	23.8 (6.5)	0.00

Notes: Mean of the effects in quarters 0 to 6 minus the mean in quarters −8 to −2, in percentage points (postings: log points × 100), with standard errors. Stacked: the design of version 1, unbalanced panel. Pre-trend p: Wald test that the effects in quarters −8 to −2 are zero.

Table C4: Posted-pay premia by occupation group

Fixed effects	Building AI	Working with AI	Postings
All postings			
Occupation × state × year	0.209 (0.013)	0.078 (0.009)	436,705
+ seniority	0.188 (0.013)	0.039 (0.008)	436,705
Firm × occupation × year	0.079 (0.004)	0.027 (0.009)	436,705
Firm × title	0.032 (0.009)	0.032 (0.015)	436,705
Computer and mathematical			
Occupation × state × year	0.173 (0.012)	−0.015 (0.013)	59,469
Firm × occupation × year	0.059 (0.005)	−0.004 (0.012)	59,469
Management and business			
Occupation × state × year	0.233 (0.016)	0.103 (0.009)	133,995
Firm × occupation × year	0.114 (0.007)	0.027 (0.011)	133,995
All other occupations			
Occupation × state × year	0.270 (0.027)	0.140 (0.020)	243,241
Firm × occupation × year	0.077 (0.017)	0.113 (0.029)	243,241

Notes: Log annual posted pay, 2023–26, on indicators for a building-AI (build) and a working-with-AI (use) duty, with fixed effects for the cells in the first column. Standard errors clustered by firm in parentheses. Postings: postings in the regression sample.

Table C5: The 30 largest rising themes of new duty lines

Theme (named by Sonnet)	Judged as	Employers 2019–22 (%)	2025–26 (%)	AI lines (%)
Integrate LLMs and AI into applications	Building AI	0.01	1.09	99
Build and operate AI platform infrastructure	Building AI	0.02	1.07	97
Leverage AI tools to work faster	Working with AI	0.01	1.00	99
Design and build AI-native workflows	Working with AI	0.01	0.96	98
Develop AI-driven product features and integrations	Building AI	0.07	0.95	96
Create and codify repeatable playbooks	Re-worded	0.22	1.06	2
Build agent platforms and tool layers	Building AI	0.03	0.87	81
Champion and drive team AI adoption	Working with AI	0.04	0.87	96
Using and adopting AI in workflows	Working with AI	0.01	0.83	99
Designing user interfaces for AI products	Building AI	0.04	0.85	97
Staying current on emerging AI capabilities	Governing/other AI	0.04	0.83	97
Building agentic AI workflows	Building AI	0.00	0.78	96
Positioning and marketing AI products	Building AI	0.03	0.80	89
Building AI agent systems and infrastructure	Building AI	0.02	0.76	100
Using AI-assisted development tools for coding	Working with AI	0.01	0.75	96
Designing and owning reliable automated workflows	Variant	0.33	1.05	5
Build and integrate LLM-based applications	Building AI	0.02	0.74	97
Ensure responsible, compliant AI governance and ethics	Governing/other AI	0.02	0.74	92
Build, launch, and sell agentic AI systems	Building AI	0.01	0.72	100
Build AI-ready data infrastructure and platforms	Building AI	0.10	0.82	93
Implement, evaluate, and improve LLM systems	Building AI	0.01	0.72	99
Lead and execute enterprise AI strategy	Governing/other AI	0.02	0.72	95
Generate and own sales pipeline	Re-worded	0.28	0.97	2
Sell AI and manage AI investments	Building AI	0.04	0.72	89
Build and operate agentic AI platforms	Building AI	0.01	0.69	98
Build backend services for AI applications	Building AI	0.03	0.70	99
Engineer AI and agentic solutions	Building AI	0.08	0.75	90
Use AI coding assistants in development	Working with AI	0.01	0.68	97
Establish governance frameworks and cadences	Re-worded	0.26	0.93	3
Use AI tools to accelerate routine work	Working with AI	0.02	0.68	96

Notes: Themes of the new duty lines of 2025–26 that rose at least threefold in the share of employers using them, ranked by the increase. Names and judgements by Sonnet 5.5. AI lines: share of the theme's lines labelled AI work.

Table C6: The 20 largest new and rising job titles

Title	Kind	Employers, 2019–22	2024–26	AI duty (%)	New lines (%)	Pay ($k)
Forward deployed engineer	rising	9	247	61	64	175
AI engineer	rising	10	168	95	70	176
GTM engineer	new	0	118	78	93	162
Senior AI engineer	rising	7	103	98	87	191
Staff product manager	rising	13	94	38	42	215
Applied AI engineer	rising	1	78	100	87	212
Founding account executive	rising	2	61	42	65	195
Senior product engineer	rising	11	59	31	57	192
AI/ML engineer	rising	3	49	100	61	162
Senior design engineer	rising	8	48	17	68	175
Deployment strategist	rising	4	44	33	71	160
Certified nursing assistant (CNA)	new	0	42	0	17	42
GTM recruiter	rising	6	41	18	25	150
Member of technical staff	rising	3	39	74	75	254
Senior forward deployed engineer	new	0	37	63	65	197
Principal product marketing manager	rising	5	37	38	37	181
Senior AI/ML engineer	rising	2	35	98	64	182
Staff electrical engineer	rising	4	34	2	41	186
Staff AI engineer	new	0	33	100	86	221
Founding product designer	rising	6	33	53	29	180

Notes: Normalised job titles with at least 10 employers in 2024–26 and none in 2019–22 (new), or at least 20 employers in 2024–26 and five times their 2019–22 count (rising), ranked by employers in 2024–26; postings for talent pools and general applications excluded. AI duty: share of the title's postings with an AI duty. New lines: share of its 2024–26 duty lines with no close counterpart among the duties of its three nearest established titles. Pay: median posted annual pay.